PORT AUTHORITY
OF NEW YORK & NEW JERSEY

Port Authority Traffic on Bridges and Tunnels

Final Progress Report

Group 6

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Executive Summary

The Port Authority of New York and New Jersey manages critical transportation facilities that keep regional movement functioning across bridges, tunnels, and connected transportation corridors. Because usage patterns can shift over time due to seasonality, vehicle mix, toll behavior, weather, holidays, special events, and pricing-related decisions, a structured analytical review was needed to better understand what affects facility demand and congestion.

This project analyzed Port Authority traffic and supporting mobility datasets to answer five core business questions related to facility usage, toll violations, busiest travel periods, 2025 congestion patterns, and future usage forecasting. The analysis combined historical traffic patterns, facility-level comparisons, toll violation summaries, vehicle-category trends, congestion indicators, and predictive modeling to create a clearer picture of how Port Authority facilities are being used and where management attention may be needed.

The final report is organized as a management-facing progress report. It begins with a data audit and structural preparation phase to explain how the datasets were validated before analysis. It then provides specific answers to each Port Authority project question, supported by numbers, patterns, model results, and dashboard visuals. The report concludes with recommendations focused on monitoring critical demand drivers, improving congestion visibility, strengthening toll violation analysis, expanding future data collection, and supporting long-term planning through forecasting.

Introduction

The Port Authority's bridges and tunnels play a central role in the movement of people, vehicles, and goods across the New York and New Jersey region. These facilities experience different levels of demand depending on time of year, day of week, vehicle type, weather conditions, holidays, toll behavior, and broader regional activity. Understanding these patterns is important because traffic pressure at one facility can affect congestion, traveler experience, operational planning, and the ability to anticipate future demand.The purpose of this project was to help Port Authority better understand the factors that influence facility usage and congestion. The analysis focused on identifying the major drivers of bridge and tunnel usage, measuring toll violation patterns, examining the busiest times throughout the year, evaluating possible congestion shifts related to 2025 pricing conditions, and forecasting facility usage beyond 2025.

The project was designed as a decision-support analysis rather than a purely technical exercise. The goal was not only to produce charts or models, but to translate the findings into clear business insights that Port Authority management can use. For that reason, the final report explains what was analyzed, why each step was necessary, what the results show, and how those results can support future monitoring and planning.

Before answering the project questions, the datasets were first audited and structurally prepared. This ensured that traffic counts, violation records, date fields, facility identifiers, vehicle categories, bus and passenger volumes, and speed measures were suitable for reliable analysis. After this preparation, the project moved into descriptive analysis, modeling, forecasting, dashboard development, and final recommendations.

Project Scope and Business Questions

This report addresses the questions provided by Port Authority for the traffic analysis of bridges and tunnels. The project focused on facilities such as the Holland Tunnel, George Washington Bridge, Lincoln Tunnel, and other outer bridge facilities included in the available data.

Project Area	**Business Question Addressed**
Facility Usage Drivers	What are the top five factors that affect the usage of bridges and tunnels by drivers?
Toll Violations	How many toll violators are there by year, month, week, and facility?
Busiest Travel Patterns	What are the busiest times throughout the year, and how is traffic affected by seasonality, toll violations, vehicle type, holidays, and related external conditions?
2025 Congestion and Pricing	Is there evidence of congestion or traffic shifts by facility in 2025, possibly connected to pricing behavior or toll avoidance?
Forecasting	What is the expected future usage of Port Authority facilities beyond 2025, based on the best available forecasting approach?

Data Audit and Structural Preparation

Purpose of the Data Audit

Before answering Port Authority's project questions, the first step was to audit the structure and quality of all raw datasets. This was necessary because the project required comparing traffic usage, toll violations, vehicle categories, bus and passenger movement, speeds, seasonality, weather, and time-based patterns across multiple facilities. If the datasets had inconsistent dates, duplicate records, missing structural fields, or unclear levels of detail, the final insights could become unreliable.

The data audit was therefore completed before any descriptive analysis, modeling, forecasting, or dashboard development. The goal was not to change the meaning of the data, but to confirm that each dataset was structurally ready for analysis. This phase focused on four areas: data completeness, duplicate records, date consistency, and dataset-level granularity. The raw datasets were preserved, and only necessary structural corrections were applied to prepare clean working files for the rest of the project.

Datasets Reviewed

Four major datasets were reviewed during the audit phase:

1. **Traffic dataset**
2. **Bus dataset**
3. **Passenger dataset**
4. **Speeds dataset**

Each dataset served a different analytical purpose. The Traffic dataset was the core operational dataset because it contained facility-level and lane-level traffic counts, toll payment categories, vehicle categories, and violation information. The Bus and Passenger datasets provided supporting transportation volume information by carrier and date range. The Speeds dataset provided facility-level mobility information that helped support the congestion-related analysis.

Traffic Dataset Audit

The Traffic dataset contained **5,383,378 rows and 30 columns**. This made it the largest and most detailed dataset used in the project. Each row represented traffic activity for a specific facility, lane, date, and time interval. This confirmed that the Traffic dataset was highly granular and operational in nature.

The audit showed that the Traffic dataset had **zero duplicate records**, which was important because duplicate rows could have inflated traffic totals, toll counts, violations, or vehicle-category volumes. The main traffic variables, including total traffic, cash payments, E-ZPass payments, toll violations, autos, small trucks, large trucks, and buses, were structurally complete.

One important finding was that the DATE field was stored as a text/object field rather than a true date field. This required conversion before the data could be used for monthly, yearly, weekly, seasonal, or day-of-week analysis. Without this correction, time-based reporting could have produced inaccurate groupings or sorting issues.

The audit also found missing values in the detailed vehicle classification columns, such as CLASS 1 through CLASS 11. These missing values were not treated as general data errors because the main traffic totals and summarized vehicle categories remained complete. The missingness was concentrated only in the detailed classification fields, which suggested inconsistent availability of granular vehicle classification rather than corruption of the core traffic data. For this reason, these values were intentionally preserved during the audit phase. This decision protected the original analytical characteristics of the dataset and avoided making unsupported assumptions about vehicle-class details.

From a business perspective, the Traffic dataset was considered reliable for answering the core Port Authority questions related to facility usage, toll violations, vehicle mix, time-based traffic patterns, and 2025 congestion analysis.

Bus Dataset Audit

The Bus dataset contained **2,784 rows and 4 columns**. The fields included start date, end date, carrier, and volume. The audit found **no duplicate records and no missing values**, which means the dataset was structurally complete.

However, the date fields were stored as text/object values and needed to be converted into proper date format. This correction was necessary because the Bus dataset used date ranges rather than a single daily timestamp. Converting both start and end dates allowed the dataset to be aligned more accurately with the broader time-based analysis.

The Bus dataset operates at a **carrier-level aggregation**, meaning each row represents bus volume by carrier within a defined date range. Because of this, it could not be directly compared to lane-level traffic records without aggregation or time alignment. This was important for the project because the final analysis needed to respect the natural level of detail in each dataset rather than forcing incompatible datasets together.

Passenger Dataset Audit

The Passenger dataset contained **2,786 rows and 4 columns**. Like the Bus dataset, it included start date, end date, carrier, and volume. The audit identified **one duplicate record** and **two incomplete rows** with missing values.

These records were removed before the analysis phase because they could have distorted summary statistics and volume-based comparisons. After cleaning, the Passenger dataset had no duplicate records and no missing values.

The date fields were also converted into proper date format to support reliable time-based analysis. Like the Bus dataset, the Passenger dataset operates at a carrier-level aggregation over defined date ranges. This made it structurally compatible with the Bus dataset after date standardization, but it still required careful handling when compared with the much more granular Traffic dataset.

Speeds Dataset Audit

The Speeds dataset contained **192 rows and 7 columns**. It included facility, free-flow speed, month-year, average speed, direction, speed difference, and facility order. The audit found **no duplicate records and no missing values**, which made it structurally clean.

The Month_Year field was stored as a text/object value and was converted to proper date format. This correction was necessary because the Speeds dataset was used in the congestion-related analysis, where facility-level monthly trends needed to be aligned with traffic and time patterns.

Unlike the Traffic dataset, which was lane-level and highly detailed, the Speeds dataset operated at a **facility-level monthly aggregation**. This means it was best suited for higher-level congestion and mobility analysis rather than detailed lane-level traffic comparisons.

Structural Corrections Applied

After completing the audit, only necessary structural corrections were applied. These corrections were made to improve analytical readiness while preserving the original meaning of the data.

The following corrections were completed:

Dataset	Structural Correction Applied	Reason
Traffic	Converted DATE to proper date format	Needed for accurate year, month, week, seasonality, and day-of-week analysis
Bus	Converted Start_Date and End_Date to proper date format	Needed to align carrier-level bus volume with time-based analysis
Passenger	Converted Start_Date and End_Date to proper date format	Needed to align passenger volume with time-based analysis
Passenger	Removed one duplicate record	Prevented double-counting or distortion of volume totals
Passenger	Removed two incomplete records	Prevented incomplete observations from affecting summary statistics
Speeds	Converted Month_Year to proper date format	Needed for monthly congestion and speed trend analysis
Traffic vehicle class fields	Missing values were preserved	Missingness reflected gaps in detailed classification, not corruption of core traffic totals

No analytical transformations were performed during this stage. The purpose was not to reshape findings or engineer insights prematurely. The purpose was to ensure that each dataset was structurally sound and could support reliable analysis in later phases.

Why These Decisions Matter

The audit confirmed that the project was built on a dependable data foundation. This matters because Port Authority's questions require decisions across several dimensions: facility, time period, vehicle type, violations, pricing, congestion, and forecasting. Each of those dimensions depends on accurate structure.

For example, if date fields were not corrected, the project could incorrectly group traffic by month, week, year, or day of week. If duplicate passenger records were left in the data, carrier-level passenger volume could be overstated. If the missing vehicle classification fields were filled without evidence, the project could create artificial patterns that did not exist in the original data.

By separating structural cleaning from analytical modeling, the project maintained a clear audit trail. The raw datasets remained unchanged, while cleaned versions were saved for descriptive analysis, modeling, forecasting, and dashboard development. This approach ensured that the final findings were based on corrected data structure without compromising the integrity of the original information.

Final Data Audit Conclusion

The audit phase confirmed that the Traffic dataset was the primary dataset for answering most of Port Authority's project questions because it contained the most detailed information on facility usage, toll violations, payment categories, vehicle categories, and time-based traffic movement. The Bus, Passenger, and Speeds datasets provided additional context for movement patterns and congestion analysis.

After structural corrections, the datasets were considered ready for the next phase of analysis. The project could then move from data validation into descriptive analysis, pattern identification, modeling, forecasting, and dashboard development with confidence that the underlying data structure was reliable.

Descriptive Analysis and Exploratory Data Evaluation.

After completing the data audit and structural preparation phase, the next step was to examine the behavior of the datasets before moving into modeling, forecasting, dashboard development, and final recommendations. This phase focused on descriptive analysis and exploratory data evaluation. The purpose was to understand how traffic, violations, payment behavior, vehicle composition, passenger activity, bus activity, and facility-level speeds behave across time and across Port Authority facilities.

This step was necessary because the project questions require more than simple totals. Port Authority is asking which factors affect usage, how many toll violators exist by time and facility, when the system is busiest, whether congestion patterns changed in 2025, and how usage may evolve in the future. To answer those questions responsibly, the data first had to be examined from several angles: system-wide behavior, temporal behavior, facility-level behavior, carrier-level transit activity, and mobility performance.

The exploratory analysis confirmed that Port Authority's traffic system is not uniform. Usage is concentrated in specific facilities, traffic volume follows seasonal and weekly patterns, E-ZPass is structurally dominant, violation behavior changed significantly after 2020, and congestion is more concentrated in certain crossings than across the entire system. These findings created the analytical foundation for the final answers to the project questions.

Traffic Dataset: Statistical Baseline

The Traffic dataset was analyzed first because it is the primary operational dataset for this project. It contains the most direct information about facility usage, toll payments, toll violations, vehicle categories, dates, time intervals, lanes, and facility identifiers. For the first descriptive overview, the analysis focused on continuous operational variables: total traffic, cash payments, E-ZPass payments, toll violations, autos, small trucks, large trucks, and buses.

Facility identifiers, lane identifiers, and time fields were not included in the first statistical summary because averages and standard deviations are not meaningful for categorical identifiers. Those fields were instead reserved for facility-level analysis, day-of-week analysis, seasonality analysis, and time-based comparisons.

At the original lane-by-time-interval level, the Traffic dataset showed strong peak-driven behavior. The average traffic per lane-time interval was approximately **270 vehicles**, while the median was approximately **199 vehicles**. Because the mean was higher than the median, the distribution was right-skewed. This means most observations were lower or moderate in size, while a smaller number of very high-volume periods pulled the average upward.

The maximum total traffic value reached approximately **3,938 vehicles** in a lane-time interval, confirming that the dataset contains extreme peak periods. These extreme values were not automatically treated as data errors because, in transportation operations, high-volume periods may reflect real demand pressure during peak travel windows. Therefore, the analysis treated these values as operationally meaningful unless later evidence suggested otherwise.

The same pattern appeared across payment and vehicle variables. E-ZPass, cash payments, violations, autos, trucks, and buses all showed right-skewed behavior, meaning activity is not evenly distributed across every facility, lane, and time interval. Instead, Port Authority traffic is shaped by concentrated periods of demand.

Business meaning:
This finding is important because congestion, enforcement pressure, staffing needs, and operational planning are usually driven by peak periods, not average periods. A simple average does not fully explain the stress placed on the system. Therefore, the analysis had to move beyond raw averages and examine time patterns, facility patterns, and volatility.

Traffic Composition and Payment Behavior

The traffic composition analysis showed that Port Authority traffic is heavily dominated by passenger vehicles and electronic tolling.

Across the full Traffic dataset, total traffic volume was approximately **1.45 billion vehicles**. E-ZPass accounted for approximately **80.96%** of total traffic, cash accounted for approximately **13.46%**, and violations represented approximately **5.58%** of total vehicles. Autos represented approximately **91.33%** of total traffic, while small trucks represented **3.33%**, large trucks represented **3.18%**, and buses represented **2.16%**.

Metric	Total Volume	Share of Total Traffic
Total Traffic	1,454,450,250	100.00%
E-ZPass	1,177,592,218	80.96%
Cash	195,756,642	13.46%
Violations	81,105,494	5.58%
Autos	1,328,400,296	91.33%
Small Trucks	48,436,522	3.33%
Large Trucks	46,205,975	3.18%
Buses	31,407,457	2.16%

This shows that the system is overwhelmingly passenger-vehicle driven. Commercial vehicles and buses are smaller shares of total traffic, but they still matter because they may be concentrated in specific facilities, time periods, or operating conditions.

The payment structure also has an important implication. Because E-ZPass represents roughly **81%** of total traffic, any future analysis related to pricing, congestion, tolling policy, or payment behavior must treat electronic tolling as the dominant structure of the system. Cash and violations remain important, but they represent smaller parts of the total payment environment.

The violation rate at the original lane-time level averaged approximately **6.96%**, but this rate was unstable at the most granular level. Some extreme rates appeared because very low traffic intervals can inflate percentages. For example, if traffic volume is very small during a lane-time interval, even a small number of violations can produce an unusually high violation percentage. For this reason, violation rates were not interpreted deeply at the raw lane-time level. The analysis moved to daily aggregation before making management-level conclusions.

Business meaning:
Electronic tolling is structurally dominant, autos drive most system demand, and violation analysis must be handled carefully because granular violation rates can be distorted by low-volume intervals.

Distribution Analysis and the Need for Aggregation

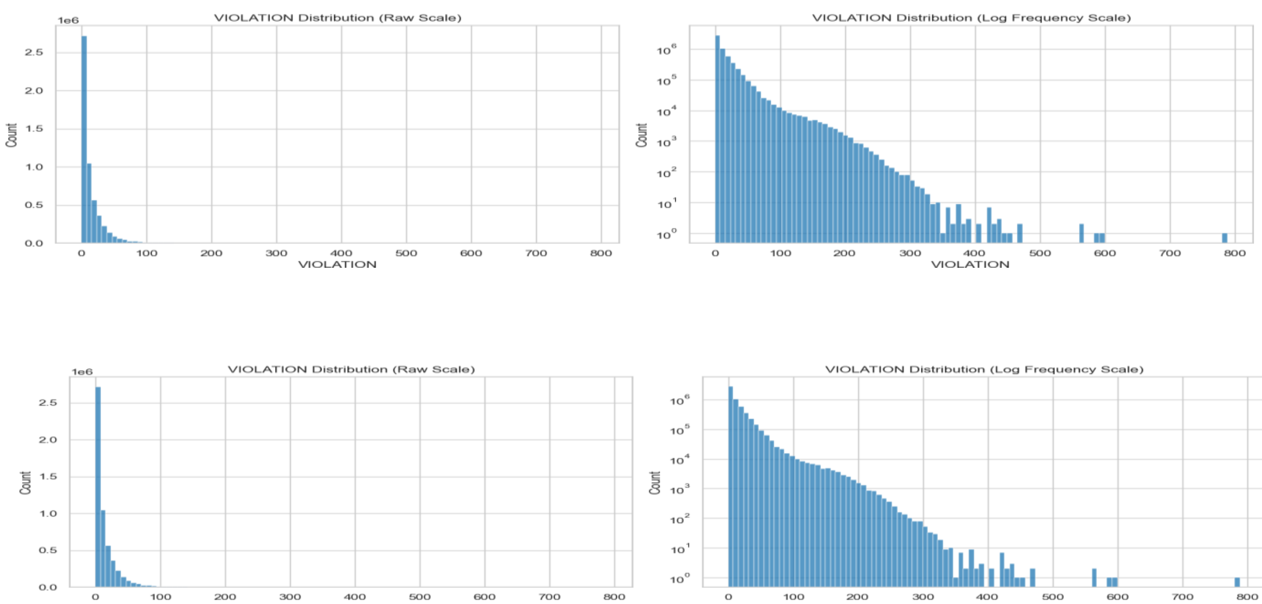
The distribution analysis confirmed that the original lane-time dataset was too volatile for direct management interpretation. Total traffic and toll violations both showed strong right-skewness. Most observations occurred at lower values, while a smaller number of observations showed very high traffic or violation counts.

This pattern does not mean the data is unreliable. In transportation systems, a long right tail is expected because traffic is often concentrated during peak commuting periods, holiday movement, event-related travel, or facility-specific demand spikes. However, this structure does mean that raw lane-time records are too noisy for answering Port Authority's broader business questions.

The analysis therefore used log-scale distribution views to make the long-tail behavior visible. This helped confirm that extreme values were sparse but operationally real. The main conclusion was that deeper analysis should not be performed at the raw lane-time level when the goal is management interpretation. Aggregation was needed to stabilize the data and reduce noise.

Business meaning:
Port Authority should not rely on raw lane-time volatility to make broad conclusions. The raw data is useful for operational detail, but daily, weekly, monthly, and facility-level aggregation are more appropriate for identifying reliable patterns.

Distribution of Total Traffic and Toll Violations at Lane-Time Level



Daily Aggregation and System-Level Stability

To create a more stable operational view, the Traffic dataset was aggregated from lane-time level to daily level. This daily aggregation combined traffic across facilities and lanes for each date and calculated total traffic, cash payments, E-ZPass payments, toll violations, violation rate, cash rate, and E-ZPass rate.

The daily-level results were much more stable and interpretable. Average daily total traffic was approximately **320,787 vehicles**, with a median of approximately **327,651 vehicles**. Average daily E-ZPass volume was approximately **259,725 vehicles**, average daily cash volume was approximately **43,175 vehicles**, and average daily violations were approximately **17,888 vehicles**.

Daily Metric	*Average Daily Value*
Total Traffic	320,787
Cash Payments	43,175
E-ZPass Payments	259,725
Toll Violations	17,888
Violation Rate	5.70%
Cash Rate	13.32%
E-ZPass Rate	80.97%

The coefficient of variation for total traffic declined from approximately **0.96** at the lane-time level to approximately **0.11** at the daily level. This confirmed that much of the raw volatility came from lane-level and time-interval variation rather than instability in the overall system.

Daily E-ZPass usage remained highly stable at approximately **81%**, with very low variability. This reinforced the finding that electronic tolling is not a temporary pattern but a stable operating feature of Port Authority traffic. Daily violation rates averaged approximately **5.70%**, which was lower and more stable than the lane-time violation rate because aggregation reduced denominator instability.

Business meaning:
Daily aggregation created a reliable base for time-series analysis. It reduced noise, stabilized violation rates, and made the data more appropriate for identifying trends, seasonality, structural changes, and forecasting patterns.

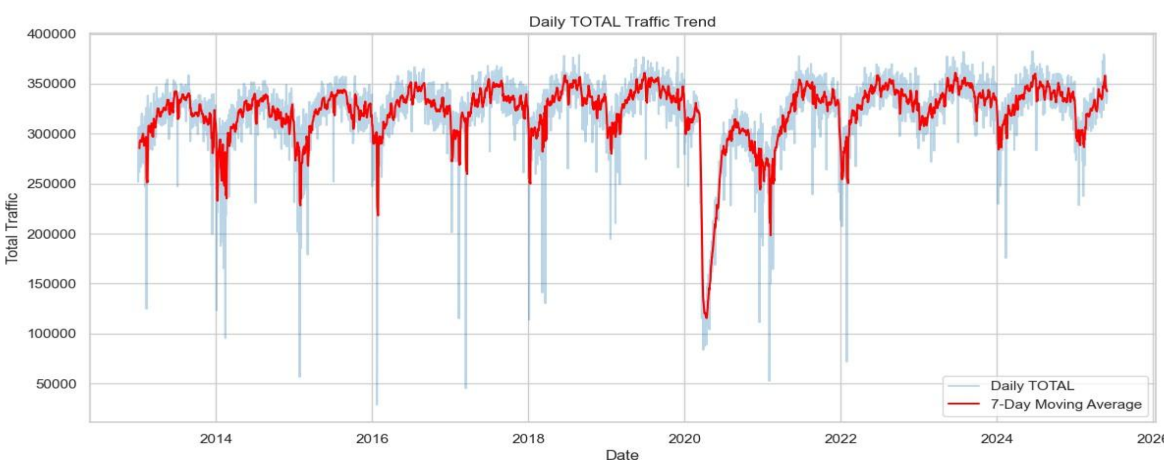
Daily Traffic Trends and Structural Break Around 2020

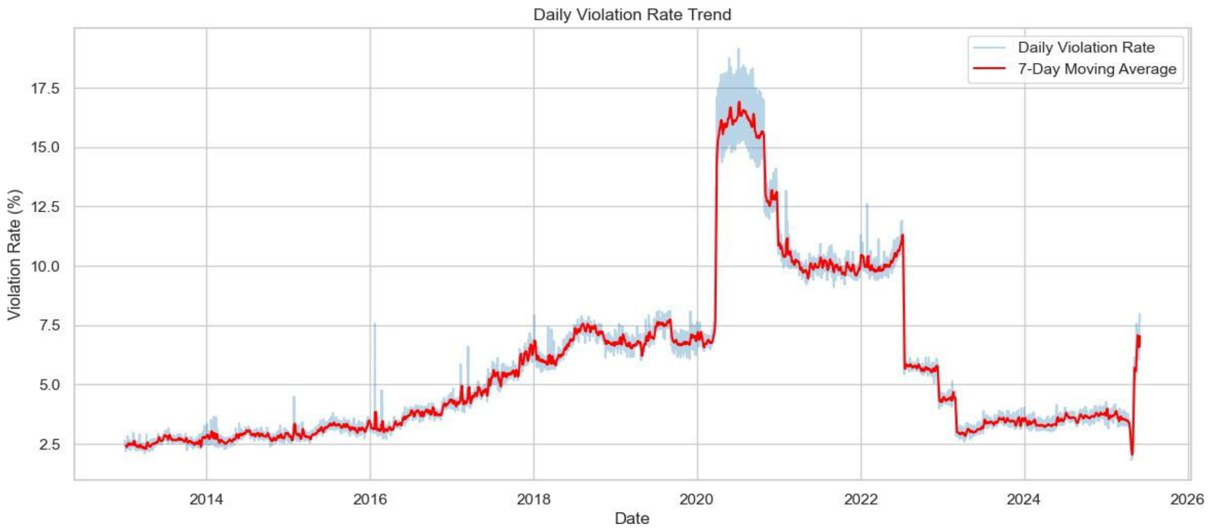
After aggregation, the daily trend analysis showed strong annual seasonality and a clear structural break around 2020. Daily traffic dropped sharply during the COVID disruption period, then gradually recovered and stabilized. This indicates that the system experienced multiple operating regimes rather than one continuous, unchanged historical pattern.

This finding is critical because it affects modeling and forecasting. If all years are treated as one identical operating environment, then the model may not properly account for the disruption and recovery period. The post-2020 system may look similar in volume to the pre-2020 system in some areas, but the underlying behavior is not identical.

Violation behavior also changed during this period. Violation rates increased gradually from 2013 to 2019, rose sharply around 2020, and later showed a structural correction. This suggests that violation behavior was influenced by more than traffic volume alone. Possible explanations may include changes in enforcement patterns, operational disruptions, tolling-system behavior, compliance behavior, or broader systemic changes. Because the dataset does not directly prove the cause, the report does not claim a single cause. Instead, it treats the change as an important structural pattern requiring management attention.

E-ZPass usage remained structurally dominant during the observed period. Although minor shifts occurred around 2020, the overall E-ZPass pattern remained stable. This means the largest behavioral change was not in electronic tolling adoption; it was in violation behavior.





Daily Trend of Total Traffic, Violation Rate, and E-ZPass Rate

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Day-of-Week Traffic Patterns

After identifying long-term trends, the analysis examined day-of-week behavior. This was necessary because daily traffic patterns can be influenced by both long-term trends and weekly cycles. Separating weekday effects helped determine whether traffic is mainly commuter-driven, weekend-driven, or consistently high across the full week.

The results showed that traffic generally increased through the week and peaked on Friday. Friday had the highest average daily traffic among the days shown, with approximately **330,534 vehicles**. Saturday and Sunday traffic remained strong, with Saturday averaging approximately **325,568 vehicles** and Sunday averaging approximately **320,721 vehicles**.

Day	Average Total Traffic	Violation Rate	E-ZPass Rate
Friday	330,533.62	5.79%	80.60%
Saturday	325,567.70	5.74%	78.09%
Sunday	320,720.71	5.73%	78.85%

The key insight is that weekend traffic remains substantial. This means the Port Authority system is not purely commuter-driven. It also supports leisure travel, family travel, shopping, tourism, airport access, events, and other non-work movement.

Violation rates remained relatively stable across weekdays, suggesting that compliance behavior does not strongly depend on day of week. However, E-ZPass usage was higher on weekdays and lower on weekends. This likely reflects a change in driver composition: weekday drivers may include more frequent commuters and registered users, while weekend traffic may include more occasional or less frequent users.

Business meaning:
Weekend traffic should not be treated as operationally minor. Port Authority should continue monitoring weekend traffic and payment composition, especially because E-ZPass usage declines on weekends even while traffic volume remains strong.

	Day_of_Week	TOTAL	Violation_Rate_%	EZPass_Rate_%
0	Monday	314,068.69	5.88	81.59
1	Tuesday	312,573.68	5.64	82.59
2	Wednesday	317,868.81	5.56	82.69
3	Thursday	324,168.39	5.57	82.40
4	Friday	330,533.62	5.79	80.60
5	Saturday	325,567.70	5.74	78.09
6	Sunday	320,720.71	5.73	78.85

Average Traffic, Violation Rate, and E-ZPass Rate by Day of Week

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Monthly Seasonality

Monthly aggregation was used to identify recurring seasonal patterns. This step helped separate short-term weekly movement from broader annual demand cycles.

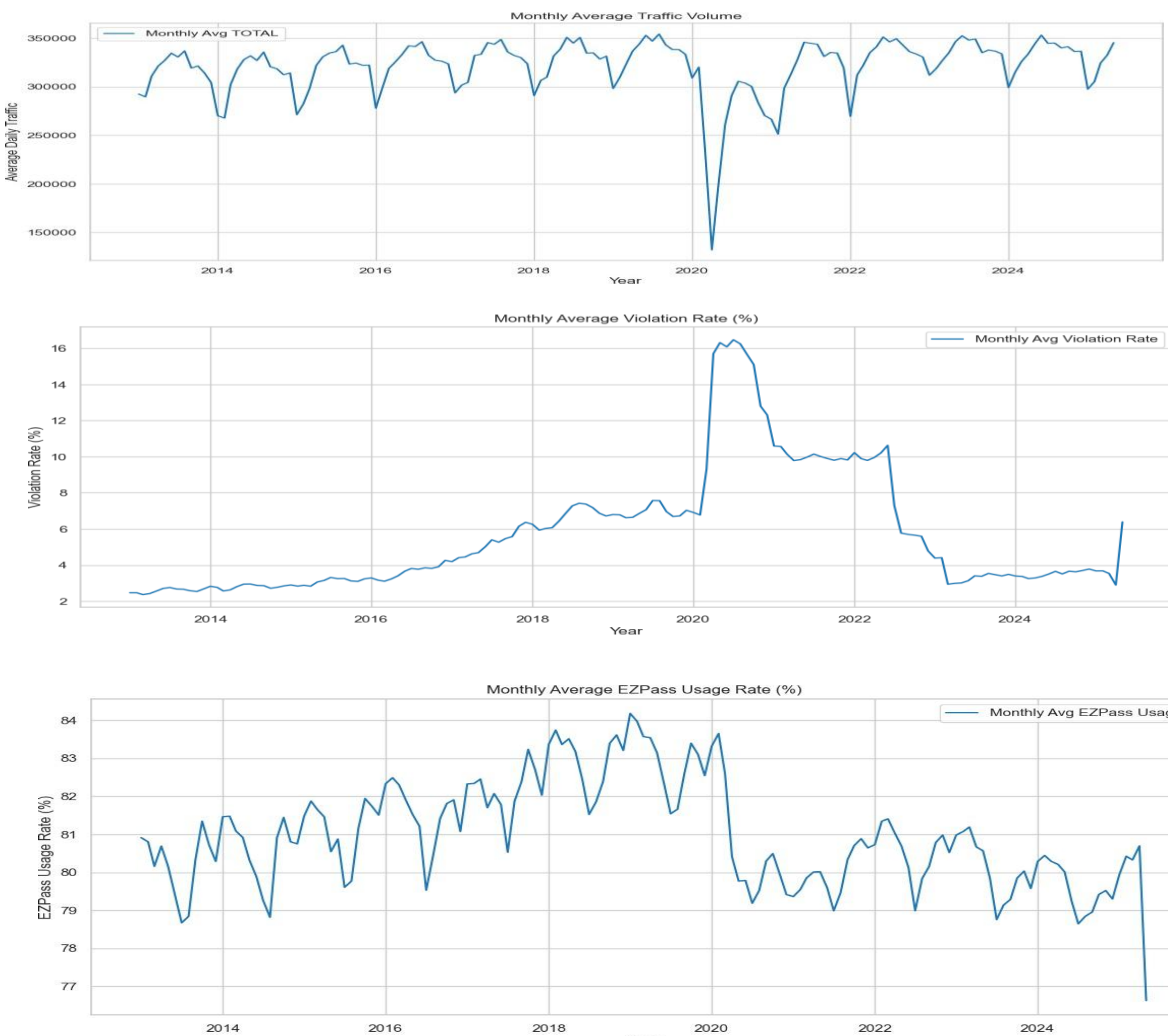
The monthly analysis showed a clear seasonal pattern in traffic volume. Excluding the 2020 disruption period, traffic generally increased through spring, peaked during late spring and summer, and declined during late fall and winter. This confirms that facility usage is seasonally driven and not only explained by commuting.

Violation rates did not follow the same seasonal rhythm. Instead, violation behavior showed a structural pattern: gradual increase before 2020, a sharp regime spike around

2020, and a downward correction after 2022. This suggests that violations are more likely influenced by systemic, operational, policy, or compliance-related factors than by seasonality alone.

E-ZPass usage showed only mild cyclical movement and remained structurally stable over time. This supports the conclusion that electronic tolling is a stable feature of the system.

Business meaning:
Traffic planning should account for seasonality, especially late spring and summer demand. However, violation management should not rely only on seasonal planning because violation behavior appears more structural than seasonal.



Monthly Trend and Seasonality of Total Traffic, Violation Rate, and E-ZPass Rate

Structural Regime Comparison: Pre-2020 vs. Post-2020

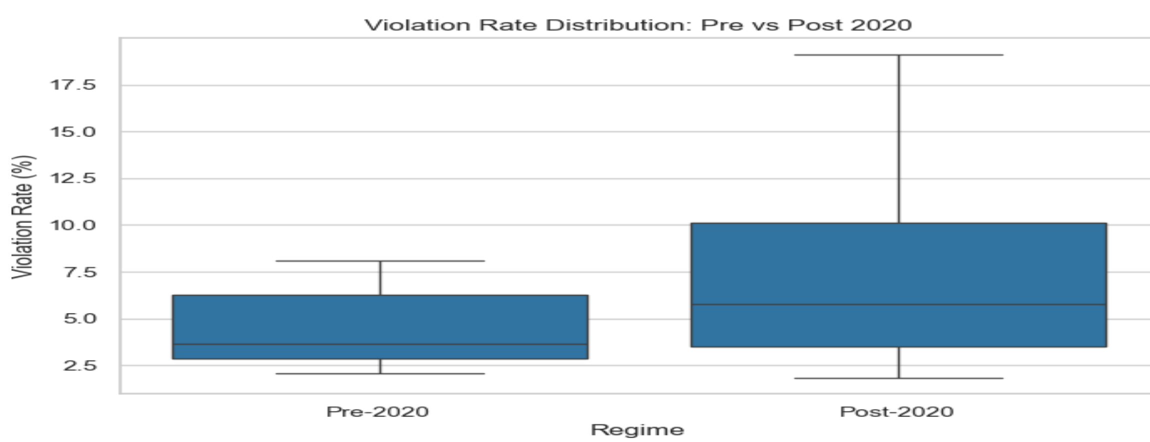
Because the daily and monthly trend analysis showed a clear break around 2020, the dataset was separated into two operating regimes: pre-2020 and post-2020. This helped evaluate whether the system returned to its earlier behavior after the disruption period or whether the post-2020 system remained structurally different.

Average daily traffic remained relatively close across the two periods. Pre-2020 average daily traffic was approximately **323,280 vehicles**, while post-2020 average daily traffic was approximately **317,567 vehicles**. This suggests that overall demand recovered close to historical levels.

However, the more important change was in violation behavior. The post-2020 period showed higher and more volatile violation rates. This means the system did not fully return to its pre-2020 behavioral structure, even if traffic volume recovered. E-ZPass usage changed only slightly, which again suggests that the main structural shift was related to compliance behavior rather than electronic tolling adoption.

Business meaning:
Port Authority should treat post-2020 violation behavior as a separate management issue. A recovery in traffic volume does not automatically mean a full recovery in compliance stability.

	TOTAL		Violation_Rate_%		EZPass_Rate_%	
	mean	std	mean	std	mean	std
Regime						
Post-2020	317,566.73	43,535.52	7.38	4.28	80.12	1.65
Pre-2020	323,279.83	28,904.56	4.41	1.78	81.63	2.85



Pre-2020 vs. Post-2020 *Comparison of Traffic, Violation Rate, and E-ZPass Rate*

Facility-Level Structural Analysis

System-wide averages can hide important facility-level differences. For that reason, the analysis moved from system-wide trends to facility-level segmentation. This step was essential because Port Authority decisions related to staffing, enforcement, congestion response, pricing monitoring, and infrastructure planning are implemented at the facility level.

The facility mapping used in the analysis is shown below.

FAC Code	Facility Name
1	Holland Tunnel
2	Lincoln Tunnel
4	Bayonne Bridge
5	Goethals Bridge
6	Outerbridge Crossing
7	George Washington Bridge Upper
8	George Washington Bridge PIP
9	George Washington Bridge Lower

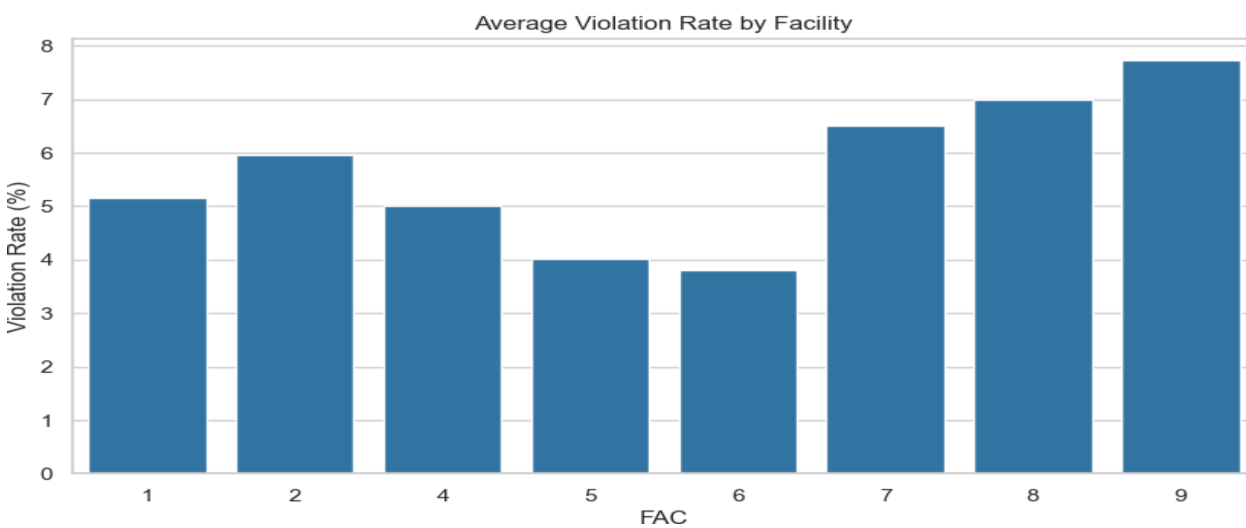
The facility-level summary showed that traffic is highly concentrated in a small number of locations. George Washington Bridge Upper had the highest average daily traffic, followed by George Washington Bridge Lower and Lincoln Tunnel.

Facility	Average Daily Traffic	Average Violation Rate	Average E-ZPass Rate
George Washington Bridge Upper	65,496.51	6.51%	76.56%
George Washington Bridge Lower	56,940.87	7.74%	80.24%
Lincoln Tunnel	50,503.50	5.95%	82.68%
Goethals Bridge	44,387.72	4.01%	82.18%
Holland Tunnel	40,982.99	5.16%	77.83%
Outerbridge Crossing	39,852.07	3.80%	86.11%
George Washington Bridge PIP	18,336.19	7.00%	86.90%
Bayonne Bridge	8,651.94	5.02%	84.49%

George Washington Bridge Upper and George Washington Bridge Lower dominate system volume, which means they can strongly influence system-wide averages. Lincoln Tunnel also carries a high average daily volume. At the same time, violation risk does not perfectly follow traffic volume. George Washington Bridge Lower had one of the highest violation rates at approximately **7.74%**, and George Washington Bridge PIP had a high violation rate of approximately **7.00%** despite lower total traffic volume.

George Washington Bridge Upper had the highest average daily traffic but a comparatively lower E-ZPass rate of approximately **76.56%**, which is below several other facilities. Holland Tunnel also had a relatively lower E-ZPass rate of approximately **77.83%**. In contrast, Outerbridge Crossing and George Washington Bridge PIP had higher E-ZPass rates above **86%**.

Business meaning:
Port Authority should not evaluate facility performance using traffic volume alone. A stronger facility view should combine three dimensions: traffic volume, violation rate, and E-ZPass rate. High-volume facilities create system pressure, but lower-volume facilities can still show elevated violation risk.



Facility-Level Average Daily Traffic, Violation Rate, and E-ZPass Rate

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Facility-Level Pre-2020 vs. Post-2020 Regime Analysis

After identifying system-wide structural changes, the analysis examined whether the post-2020 shift affected all facilities equally or whether it was concentrated in specific locations.

The results showed that violation rates increased sharply at several facilities after 2020, especially within George Washington Bridge-related facilities and Lincoln Tunnel.

Facility	Pre-2020 Violation Rate	Post-2020 Violation Rate	Direction of Change
Holland Tunnel	3.96%	6.71%	Increased
Lincoln Tunnel	4.27%	8.12%	Increased
Bayonne Bridge	5.71%	4.14%	Decreased
Goethals Bridge	4.15%	3.84%	Slightly decreased
Outerbridge Crossing	4.64%	2.70%	Decreased
George Washington Bridge Upper	4.22%	9.48%	Increased sharply
George Washington Bridge PIP	4.17%	14.83%	Increased sharply
George Washington Bridge Lower	5.05%	11.22%	Increased sharply

The most significant increases occurred at George Washington Bridge PIP, George Washington Bridge Lower, George Washington Bridge Upper, Lincoln Tunnel, and Holland Tunnel. This shows that the post-2020 compliance shift was not evenly distributed across all facilities.

Traffic volumes remained broadly stable across many facilities, but violation rates changed much more sharply. This suggests that the structural break was not only a demand issue. It was more strongly connected to compliance behavior, operational conditions, or facility-specific changes.

Business meaning:
Port Authority should prioritize facility-level violation monitoring, especially at George Washington Bridge PIP, George Washington Bridge Lower, George Washington Bridge Upper, Lincoln Tunnel, and Holland Tunnel. These locations showed meaningful post-2020 increases and may require closer review in enforcement, toll compliance, operational processes, or customer behavior.

FAC	REGIME	TOTAL	VIOLATION_RATE_%	EZPASS_RATE_%
1	Post-2020	39,301.01	6.71	75.92
1	Pre-2020	42,284.62	3.96	79.31
2	Post-2020	48,923.13	8.12	81.93
2	Pre-2020	51,726.49	4.27	83.25
4	Post-2020	10,259.07	4.14	81.59
4	Pre-2020	7,384.94	5.71	86.78
5	Post-2020	46,986.88	3.84	82.11
5	Pre-2020	42,376.32	4.15	82.23
6	Post-2020	39,045.78	2.70	86.86
6	Pre-2020	40,476.03	4.64	85.53
7	Post-2020	68,936.06	9.48	76.10
7	Pre-2020	62,834.77	4.22	76.91
8	Post-2020	15,800.12	14.83	85.02
8	Pre-2020	19,250.01	4.17	87.58
9	Post-2020	56,757.92	11.22	79.02
9	**PRE-2020**	**57,082.45**	**5.05**	81.18

Pre-2020 vs. Post-2020 Violation Rate and EZPASS Rate by Facility

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Facility-Level Variability and Operational Risk

Average traffic and average violation rates show typical behavior, but variability shows predictability. A facility with moderate average traffic but high volatility may still require management attention because it is harder to plan for.

The variability analysis showed that traffic volume was relatively stable across major facilities, while violation rates were much more volatile. This means operational risk is more strongly tied to compliance instability than to traffic instability.

Facility	Traffic CV	Violation CV	Interpretation
Holland Tunnel	0.13	0.93	Stable traffic but highly unstable violation behavior
George Washington Bridge Upper	0.16	0.85	High-volume facility with elevated violation volatility
Bayonne Bridge	0.32	0.80	Smaller facility with higher proportional traffic and violation volatility

George Washington Bridge Lower	0.14	0.80	High-volume facility with unstable violation behavior
George Washington Bridge PIP	0.16	0.72	Lower-volume facility with elevated violation risk
Lincoln Tunnel	0.14	0.71	Stable traffic with moderate-to-high violation variability
Outerbridge Crossing	0.13	0.44	Relatively stable traffic and violation behavior
Goethals Bridge	0.15	0.35	Relatively stable violation behavior

Holland Tunnel had the highest violation variability, while George Washington Bridge Upper and George Washington Bridge Lower combined high traffic with elevated violation volatility. Bayonne Bridge had the highest traffic coefficient of variation, which is expected because smaller-volume facilities can show higher proportional movement even when absolute traffic changes are not as large.

Business meaning:
Operational risk should be measured using both level and volatility. Facilities with high traffic and high violation variability should receive priority attention because they combine scale with unpredictability. Facilities with lower traffic but high proportional volatility should also be monitored because they may require different planning assumptions.

Correlation and Relationship Analysis

After examining trends, seasonality, regime shifts, and facility-level behavior, the analysis evaluated whether key variables were strongly related to one another.

The correlation analysis showed that total traffic and violation rate had a moderate negative relationship of approximately **-0.33**. E-ZPass rate had near-zero correlation with both total traffic and violation rate.

Relationship	Correlation	Interpretation
Total Traffic vs. Violation Rate	-0.33	Higher traffic does not automatically create higher violation rates
Total Traffic vs. E-ZPass Rate	-0.03	Traffic volume is not meaningfully related to E-ZPass share
Violation Rate vs. E-ZPass Rate	-0.03	Violation behavior is not mechanically explained by E-ZPass adoption

This finding prevents an oversimplified conclusion. It would be easy to assume that higher traffic automatically causes higher violation rates, but the data does not support that as a direct relationship. Violation behavior appears to be influenced by structural, operational, regime-specific, or facility-specific factors rather than traffic volume alone.

The scatter analysis also showed clustering, suggesting that the relationship between traffic volume and violation rate is shaped by structural regime changes rather than one continuous linear pattern. This supports the earlier conclusion that post-2020 behavior should be treated separately.

Business meaning:
Port Authority should not design violation strategies based only on traffic volume. Compliance monitoring should also include facility-level differences, post-2020 regime behavior, anomaly periods, and payment-pattern changes.

Scatter Plot of Total Traffic vs. Violation Rate

Outlier and Anomaly Detection

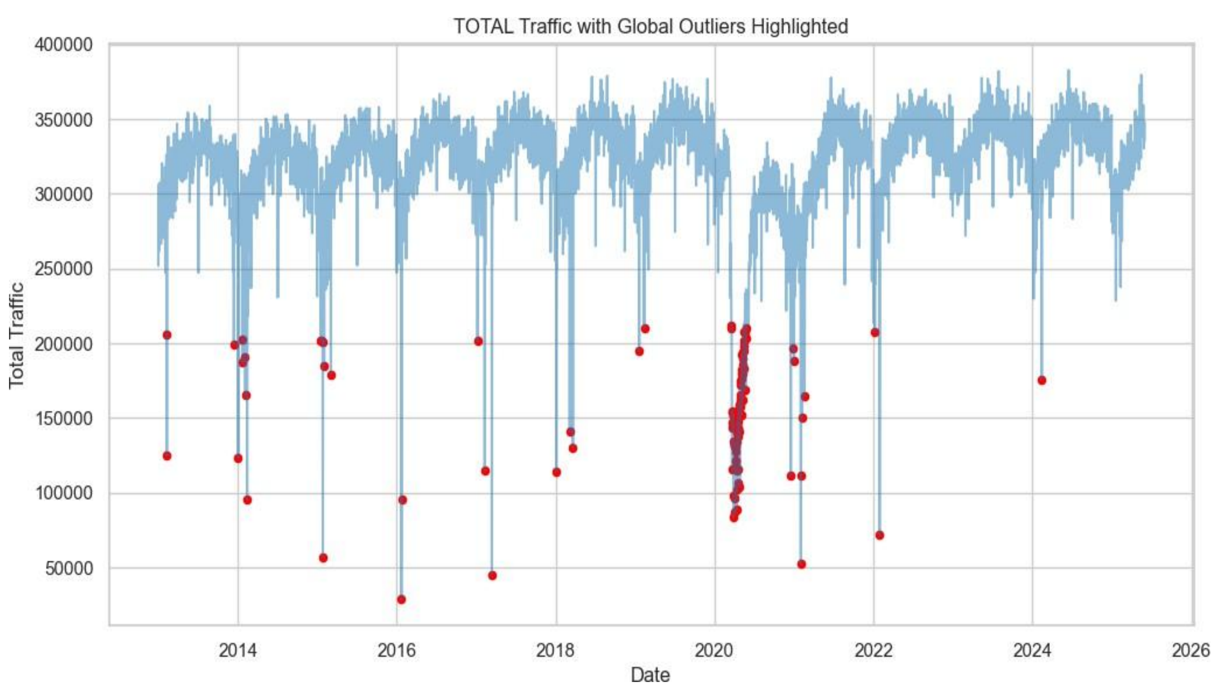
The exploratory analysis also examined whether extreme traffic days were random events or part of larger structural disruptions.

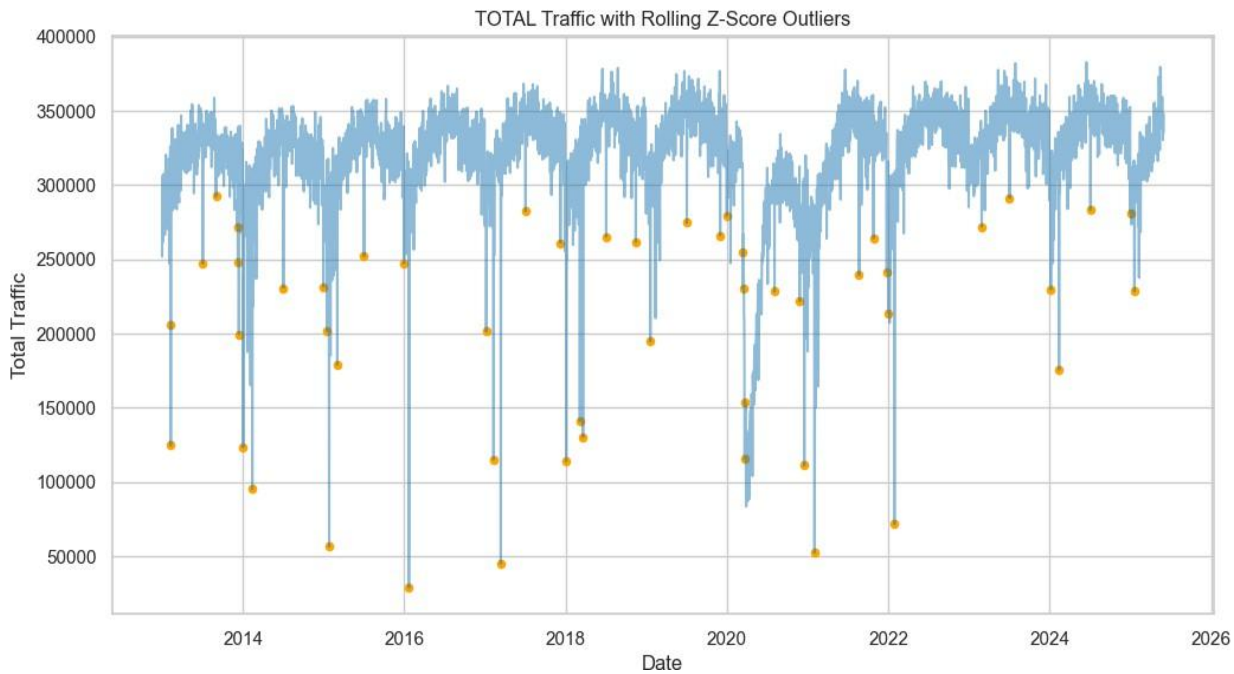
Using a global Z-score method, **84 days** were identified as extreme deviations from long-run traffic levels. Most of these extreme observations were concentrated around the 2020–2021 period, indicating that they were connected to a major structural disruption rather than normal daily variation.

A rolling Z-score method was also used to identify localized anomalies relative to recent historical behavior. This method identified **51 localized anomalies**. Many of these were also concentrated around the 2020–2021 disruption period, but the rolling method also helped identify shorter-term shocks outside the main disruption window.

These outlier and anomaly periods were not removed during exploratory analysis. This was intentional. Extreme traffic days may represent real operational events, not errors. Removing them too early could erase important information about disruption behavior. However, leaving them unflagged could distort modeling and forecasting. Therefore, these periods were identified and preserved for careful treatment in later phases.

Business meaning:
Port Authority should maintain anomaly flags in future analytical systems. Extreme days should be classified and reviewed instead of automatically removed. Possible explanations may include major disruptions, severe weather, special events, reporting issues, policy changes, or unusual travel behavior.





Formal Traffic Data Quality Confirmation

After the exploratory analysis, the daily traffic dataset was reviewed again for missing values and duplicate records. The core operational variables, including date, total traffic, cash, E-ZPass, violations, violation rate, cash rate, and E-ZPass rate, had no missing values at the daily level. Duplicate detection also returned zero duplicate daily records.

The only missing values appeared in rolling statistics such as rolling mean, rolling standard deviation, and rolling Z-score. These missing values were expected because rolling calculations require a minimum number of prior observations. For example, a 30-day rolling window cannot produce values for the first 29 observations. This does not indicate a raw data quality problem.

Business meaning:
The daily traffic dataset was structurally reliable for time-series analysis, facility segmentation, anomaly detection, and preparation for later modeling.

Passenger Dataset Analysis

The Passenger dataset operates at a weekly carrier-level aggregation. Each row represents passenger volume for a specific carrier during a defined weekly period. This dataset provides supporting context for commuter and transit-related demand patterns.

Passenger volume was highly right-skewed. The mean weekly passenger volume was approximately **6,178**, while the median was approximately **796**. This large gap shows that a small number of high-volume carrier-weeks strongly influence the average. Passenger volume ranged from **0** to **90,225**, confirming substantial variability across carriers and time periods.

The weekly trend showed a clear upward structural movement from 2021 through 2025. This indicates sustained recovery and growth in passenger demand. However, several abrupt drops to near-zero levels were also observed. These drops may reflect reporting gaps, temporary service interruptions, operational disruptions, or structural breaks in the data. Because the cause is not confirmed directly in the dataset, they are treated as items requiring review rather than assigned a specific explanation.

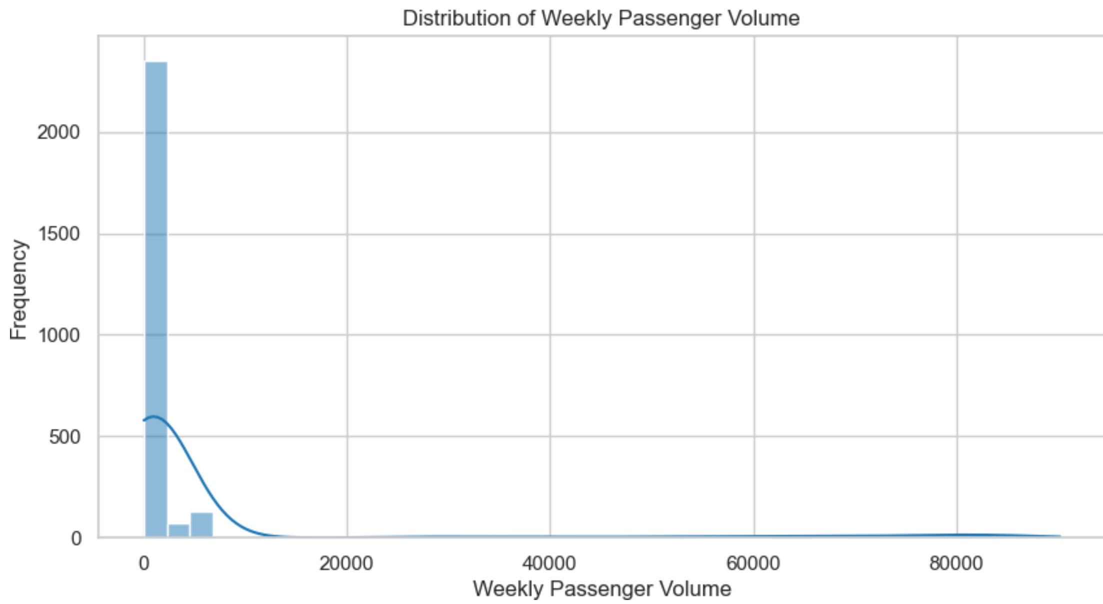
Carrier-level analysis showed strong concentration. NJTransit accounted for approximately **12.99 million** total passenger volume, making it the dominant passenger carrier in the dataset. The next largest recorded NJ Transit entry had approximately **1.45 million**, but the presence of both "NJTransit" and "NJ Transit" indicates a naming inconsistency that should be standardized before integration. Other carriers operated at much smaller scales.

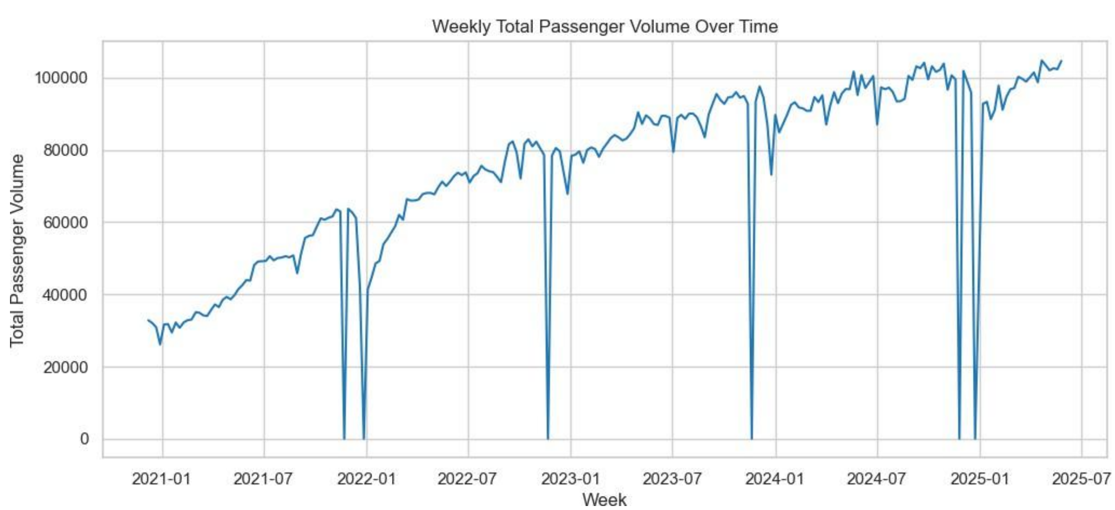
Passenger Carrier	Total Volume	Mean Volume	Observation
NJTransit	12,985,016	60,395.42	Dominant passenger carrier
NJ Transit	1,452,175	85,422.06	Same carrier name structure likely needs standardization
Coach USA	972,252	4,190.74	Largest non-NJ Transit passenger carrier
HCEE - Community	330,976	1,426.62	Moderate passenger activity
Greyhound	320,417.40	1,381.11	Moderate passenger activity
Academy	245,524.20	1,076.86	Naming fragmentation appears in dataset
Lakeland	240,602	1,037.08	Moderate passenger activity

Several carriers showed very low volumes or fragmented naming patterns, including variations of C&J Bus Lines and Academy. This confirms that carrier name standardization is necessary before merging passenger data with bus data.

Business meaning:

Passenger demand is heavily shaped by NJ Transit activity. System-wide passenger trends should therefore be interpreted carefully because one dominant carrier can drive the overall pattern. Carrier standardization is required before any final integrated transit analysis.





Bus Dataset Analysis

The Bus dataset also operates at a weekly carrier-level aggregation. Each row represents bus volume for a specific carrier during a defined weekly period. Because its structure is similar to the Passenger dataset, it is useful for comparing bus activity with passenger volume after carrier names are standardized.

Weekly bus volume was strongly right-skewed. The mean weekly bus volume was approximately **222**, while the median was approximately **29**. This shows that most carrier-week observations had low bus volume, while a smaller number of high-volume observations pulled the average upward. Bus volume ranged from **0** to **2,577**, confirming uneven operational intensity across carriers and time.

The weekly bus trend showed steady growth from 2021 through 2025. Compared with passenger volume, bus volume appeared more stable overall, although several abrupt drops to near-zero levels were still observed. These drops should be reviewed before integration to determine whether they represent true service interruptions, reporting gaps, or other structural issues.

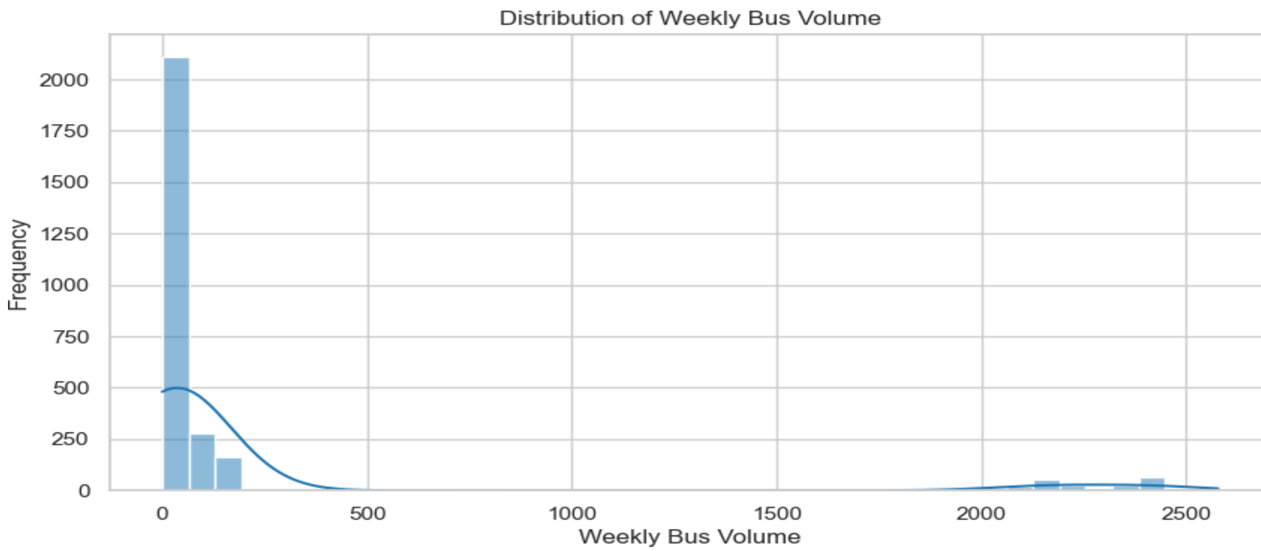
Carrier-level analysis showed that NJ Transit dominated bus volume, with approximately **515,189** total bus volume. Coach USA was the next largest carrier, but at a much smaller scale.

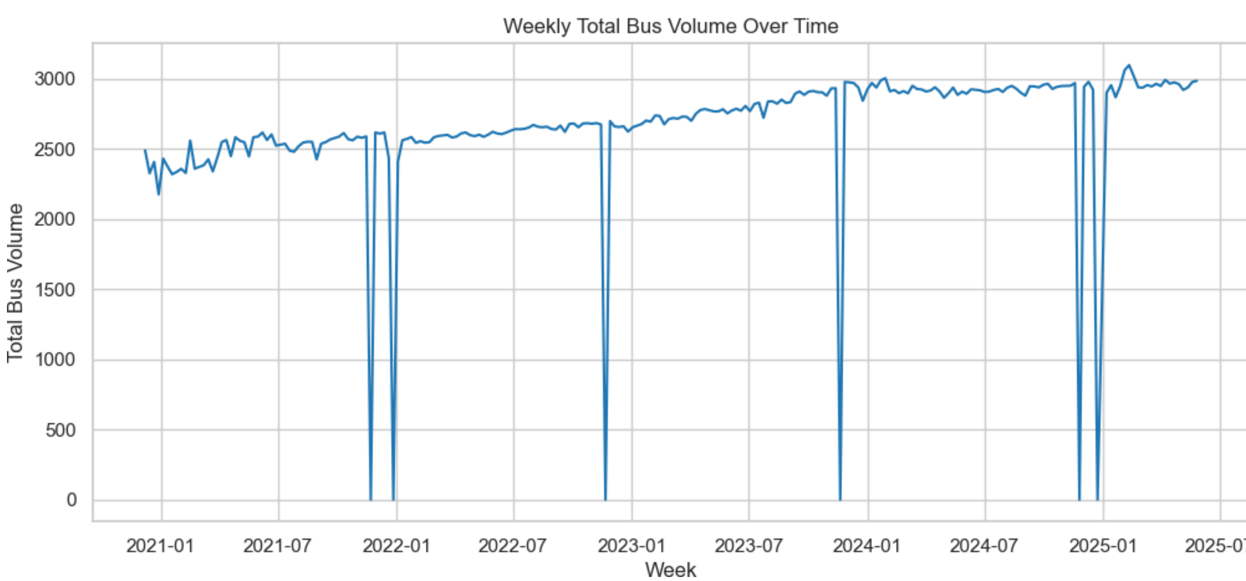
Bus Carrier	**Total Volume**	**Mean Volume**	**Observation**
NJ Transit	515,189	2,220.64	Dominant bus carrier
Coach USA	32,892	141.78	Largest non-NJ Transit bus carrier
HCEE - Community	18,901	81.47	Moderate bus activity
Greyhound	12,029	51.85	Moderate bus activity
Academy	8,419	36.93	Naming fragmentation appears in dataset
Lakeland	7,180	30.95	Lower-volume carrier
Peter Pan_Bonanza	6,293.40	27.13	Lower-volume carrier

The Bus and Passenger datasets are structurally compatible because both operate at carrier-week level. However, carrier name standardization is required before they can be reliably compared or merged.

Business meaning:

Bus activity, like passenger activity, is concentrated in a small number of carriers. NJ Transit strongly influences system-level bus trends. Before using bus and passenger data together, carrier names must be standardized to avoid splitting one carrier into multiple records.





Speeds Dataset Analysis

The Speeds dataset operates at a facility-by-direction-by-month level. Unlike the Traffic, Bus, and Passenger datasets, which measure volume, the Speeds dataset measures mobility performance. It includes facility, free-flow speed, month, average observed speed, travel direction, speed delta, and facility order.

The facility abbreviations used in the Speeds dataset were interpreted as follows:

Speeds Code	Facility Name
BB	Bayonne Bridge
GB	Goethals Bridge
GWB	George Washington Bridge
HT	Holland Tunnel
LT	Lincoln Tunnel
OBX	Outerbridge Crossing

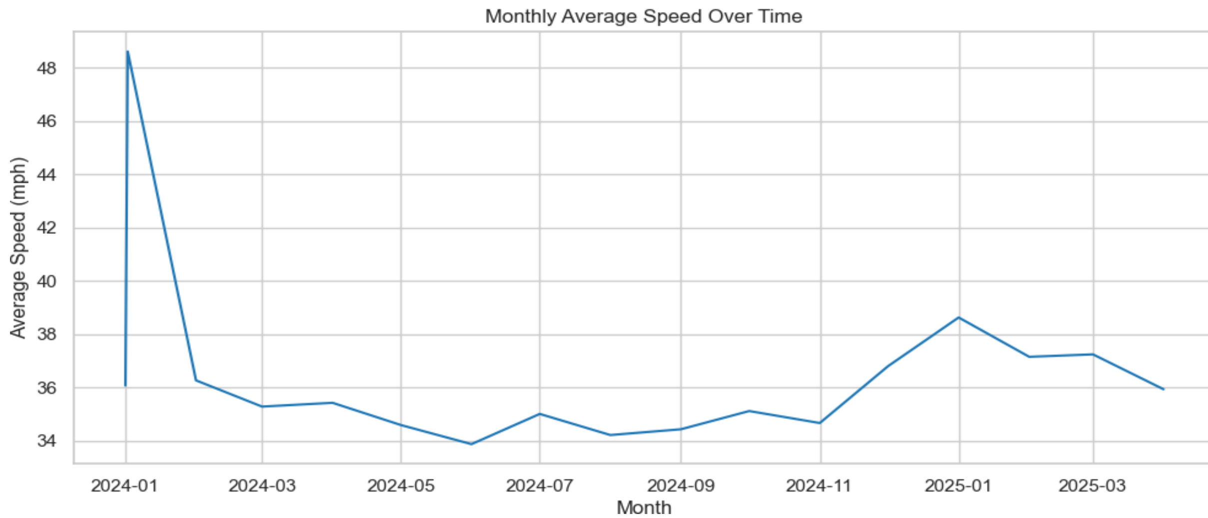
The average observed speed was approximately **35.73 mph**, while the average free-flow speed was approximately **42.02 mph**. The reported average speed delta was approximately **-6.52 mph**, indicating that observed speeds were generally below free-flow conditions. Average observed speeds ranged from approximately **13.20 mph** to **49.50 mph**, showing meaningful variation across facilities, directions, and months.

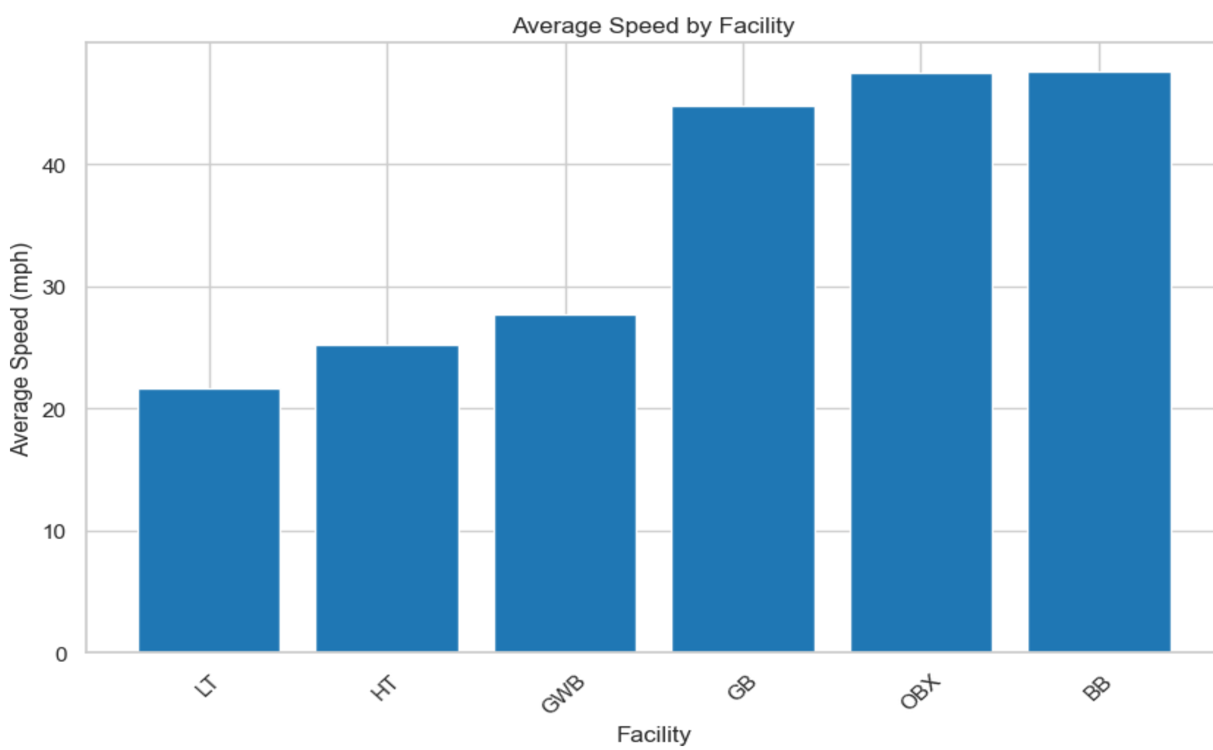
Monthly speed trends appeared relatively stable during the observed period. After an early spike, average speeds generally fluctuated within a narrow band between approximately **34 and 38 mph**. No clear long-term deterioration was observed at the system level.

However, facility-level analysis showed that congestion is not evenly distributed. Lincoln Tunnel, Holland Tunnel, and George Washington Bridge showed substantially lower average speeds, approximately in the **21–28 mph** range. In contrast, Outerbridge Crossing, Bayonne Bridge, and Goethals Bridge had average speeds closer to **45–48 mph**, suggesting conditions closer to free-flow movement.

Business meaning:
Congestion is facility-specific, not uniform across the system. System-wide average speed can hide severe congestion at specific corridors. Port Authority should evaluate

mobility performance by facility and direction, especially for Lincoln Tunnel, Holland Tunnel, and George Washington Bridge.





Cross-Dataset Alignment and Integration Blueprint

The exploratory analysis also clarified how the four datasets can be aligned for later analysis. This was important because each dataset operates at a different level of detail.

Dataset	Granularity	Main Metrics	Grouping Variables
Traffic	Daily after aggregation	Total traffic, violations, E-ZPass, cash	Date, facility
Passenger	Weekly	Passenger volume	Week, carrier
Bus	Weekly	Bus volume	Week, carrier
Speeds	Monthly	Average speed, free-flow speed, speed delta	Month, facility

The Traffic and Speeds datasets can be connected through facility and month after aggregating daily traffic to monthly facility-level totals. This allows the project to compare traffic volume, violation behavior, and speed performance.

The Passenger and Bus datasets can be connected through week and carrier after standardizing carrier names. This allows later analysis of passenger-to-bus relationships, service intensity, and carrier-level concentration.

The Traffic dataset can also be compared indirectly with Passenger and Bus datasets by aggregating traffic to weekly level. This allows broader comparison between vehicle traffic and transit activity over time.

Weather can be incorporated directly with Traffic at the daily level and then aggregated to weekly or monthly level when comparing with Passenger, Bus, or Speeds data. Weather is relevant because temperature, precipitation, snow, and severe weather conditions can affect traffic volume, bus activity, passenger movement, and average speeds.

Business meaning:
The datasets should not be forced into one raw-level merge. They must first be harmonized to compatible time levels. This protects the analysis from duplication, distorted totals, and incorrect conclusions.

Final EDA Conclusion

The descriptive analysis and exploratory data evaluation produced several important findings that shaped the rest of the project.

First, the Traffic dataset is highly peak-driven at the raw lane-time level. This makes raw records useful for operational detail but too volatile for management-level conclusions. Daily aggregation created a more stable view of system behavior and became the appropriate base for trend, seasonality, and facility-level analysis.

Second, E-ZPass is structurally dominant, representing approximately **81%** of total traffic. This means electronic tolling is central to Port Authority's payment environment and must be considered in any future pricing, congestion, or compliance analysis.

Third, autos dominate the vehicle mix, representing more than **91%** of total traffic. Trucks and buses represent smaller shares, but their higher volatility means they may still matter during specific facilities, periods, or operational conditions.

Fourth, traffic volume shows clear annual seasonality. Demand generally rises through spring, peaks in late spring and summer, and declines during late fall and winter. This finding supports the need for seasonal planning.

Fifth, violation behavior changed structurally after 2020. The post-2020 period showed higher and more volatile violation rates, especially at George Washington Bridge Upper, George Washington Bridge PIP, George Washington Bridge Lower, Lincoln Tunnel, and Holland Tunnel. This suggests that violation behavior should be monitored as a separate operational risk rather than assumed to move directly with traffic volume.

Sixth, facility-level differences are substantial. George Washington Bridge Upper, George Washington Bridge Lower, and Lincoln Tunnel drive large portions of traffic volume, while certain facilities show higher violation rates or higher violation volatility. This confirms that facility-level monitoring is necessary.

Seventh, the Speeds dataset shows that congestion is concentrated in specific crossings. Lincoln Tunnel, Holland Tunnel, and George Washington Bridge show lower speed performance, while Outerbridge Crossing, Bayonne Bridge, and Goethals Bridge operate closer to free-flow conditions.

Finally, the datasets are structurally compatible for integration, but only after temporal harmonization, facility-name standardization, carrier-name normalization, and careful review of outliers and zero-volume periods. These steps protect the integrity of later modeling, forecasting, dashboard development, and final recommendations.

Overall, the EDA phase confirmed that Port Authority's traffic and mobility environment is structured, concentrated, seasonal, and facility-specific. These findings provide the foundation for answering the project questions in a management-ready way.

Data Cleaning, Grain Alignment, and Dataset Integration.

After completing the data audit and exploratory analysis, the next step was to prepare the datasets for integration. At this stage, the datasets were structurally valid, but they were not yet ready to be merged because they existed at different levels of detail. The Traffic dataset was recorded at a lane, date, and time-interval level. The Passenger and Bus datasets were recorded at a carrier and weekly date-range level. The Speeds dataset was recorded at a facility and monthly level. The Weather dataset was recorded at a daily level.

Because these datasets did not share the same analytical grain, direct merging would have created serious data issues. For example, joining lane-level traffic directly with weekly passenger data could artificially duplicate passenger volumes across many traffic records. Joining monthly speed values directly to daily traffic without proper alignment could also create mismatched or repeated values. Therefore, before any integration, the project first defined a single analytical grain that would control all cleaning, aggregation, and merging decisions.

Analytical Grain Selection

The final analytical grain selected for the master dataset was:

Facility × Date

This means that each row in the final integrated dataset represents one facility on one specific date.

This grain was selected because it provided the best balance between operational detail and analytical stability. The Traffic dataset could be aggregated from lane-time level to daily facility totals without losing meaningful volume information. Weather data naturally aligned to daily frequency. Passenger and Bus datasets could be transformed from weekly records into daily representations. Speed data could be connected at the monthly facility level if facility and month keys aligned correctly.

The Facility × Date grain also reduced unnecessary complexity. The original lane-time traffic file contained millions of rows, which was useful for operational detail but too granular for management-level modeling and dashboard reporting. Aggregating to daily facility-level records preserved facility differences while making the dataset stable enough for modeling, forecasting, and business interpretation.

Interpretation:
The grain decision was one of the most important design choices in the project. It prevented row inflation, protected the accuracy of traffic totals, and ensured that every future calculation had a clear business meaning. Instead of mixing datasets at incompatible levels, the project created a consistent one-row-per-facility-per-day structure.

Traffic Dataset Cleaning and Aggregation

The Traffic dataset was treated as the backbone of the project because it contained the main operational measures needed to answer Port Authority's questions: total usage, toll payments, toll violations, vehicle categories, dates, time intervals, lanes, and facility identifiers.

The original Traffic dataset operated at the following grain:

Facility × Lane × Date × Time Interval

For the final master dataset, it was transformed into:

Facility × Date

Before aggregation, the Traffic dataset was checked for duplicate behavior at the operational grain. A total of **41,101 rows** shared the same facility, lane, date, and time combination. However, these were not treated as simple duplicate records because the values within those grouped records varied. This means they represented multiple recorded entries within the same operational time slot rather than accidental copy-paste duplicates. Because the records contained distinct traffic values, they were preserved. Aggregation to the facility-date level then consolidated them naturally without deleting legitimate traffic counts.

The Traffic dataset was also checked for internal consistency between total traffic and payment components. Specifically, the analysis tested whether:

Total Traffic = Cash + E-ZPass + Violations

Most records were consistent. A limited number of discrepancies were found, with most differences being very small, typically plus or minus one or two vehicles. Only **24 rows** showed larger differences greater than 10 vehicles. These differences were documented but not manually altered. The reason is that total traffic was fully consistent with vehicle category totals across all rows, making TOTAL the most reliable measure of overall facility usage. Therefore, TOTAL was treated as the authoritative traffic volume measure.

Interpretation:
This was a conservative and defensible cleaning decision. Instead of forcing payment fields to match total traffic, the project preserved the original reported values and used the most internally consistent measure for traffic volume. This avoids creating artificial corrections that could distort the operational data.

After validation, all traffic volume fields were aggregated to the Facility × Date level. The final traffic_daily dataset contained **35,168 rows and 10 columns**. A validation check confirmed that total traffic volume was fully preserved during aggregation:

Measure	Value
Original Total Traffic Sum	1,454,450,250
Aggregated Total Traffic Sum	1,454,450,250

This confirmed that the aggregation process did not lose or inflate traffic volume. No missing values were introduced during aggregation, and the daily facility-level traffic table became the backbone dataset for integration.

Interpretation:
This step ensured that the project could move forward with a stable, reliable traffic foundation. The final daily traffic table preserved total system volume, maintained facility-level detail, and removed unnecessary lane-time noise from the modeling dataset.

Facility Mapping Used for Interpretation

The Traffic dataset used numeric facility codes. For management reporting, those codes were converted into readable facility names so that the results would be understandable to Port Authority decision-makers.

Facility Code	Facility Name
1	Holland Tunnel
2	Lincoln Tunnel
4	Bayonne Bridge
5	Goethals Bridge
6	Outerbridge Crossing
7	George Washington Bridge Upper
8	George Washington Bridge PIP
9	George Washington Bridge Lower

This mapping was important because facility-level results are central to the project. Reporting only numeric facility codes would make the analysis harder to interpret and could create confusion when discussing congestion, violations, pricing effects, and future usage patterns.

Weather Dataset Cleaning and Integration

After the Traffic dataset was finalized at the Facility × Date level, the Weather dataset was prepared because weather naturally operates at a daily frequency. This made it structurally compatible with the selected master grain.

The Weather dataset contained **4,534 rows and 7 columns**. It included the following fields: station, date, average wind speed, precipitation, snowfall, maximum temperature, and minimum temperature. The date range matched the traffic dataset, covering **January 1, 2013 through May 31, 2025**.

The Weather dataset was cleaned by converting the date field into proper date format and converting weather measurements into standard units. Maximum and minimum temperatures were converted from tenths of degrees Celsius into degrees Celsius. Precipitation and average wind speed were also converted from tenths into standard units. An average daily temperature field was created using the maximum and minimum temperature values.

The Weather dataset had no missing values in date, precipitation, snow, maximum temperature, or minimum temperature. Average wind speed had **225 missing values**, representing approximately **4.96%** of the Weather dataset. These missing wind values were not dropped because the dates themselves were valid and wind speed was not a merge key. Dropping those rows would have removed valid traffic dates from the analysis. Instead, the missingness was documented and handled during final dataset preparation.

The Weather data was then merged with the daily facility-level traffic backbone using the DATE field. After the merge, the master dataset contained **35,168 rows**, preserving the Facility × Date structure.

Interpretation:
Weather was included because it is an external factor that can influence travel behavior, congestion, safety conditions, and daily traffic variability. Since it does not depend on traffic volume, it strengthens the project by adding an exogenous explanatory layer. Integrating it at the daily level allowed the project to evaluate whether weather conditions helped explain changes in usage, violations, or congestion-related patterns.

Passenger and Bus Dataset Alignment

The Passenger and Bus datasets were structurally different from the Traffic dataset because they were recorded at a weekly carrier-level grain. Each row represented a carrier over a start and end date range rather than one facility on one day.

To align these datasets with the master Facility × Date grain, the weekly records were transformed into daily representations. This allowed passenger and bus volume information to be connected to the daily traffic backbone. After transformation, the Passenger and Bus datasets were aggregated to date-level totals and merged into the master dataset using left joins.

A left-join strategy was used because the traffic dataset was the backbone. This ensured that every facility-date traffic record remained in the master dataset even if passenger or bus volume was not available for that date. Missing passenger and bus values after the merge were filled with zero, meaning there was no recorded passenger or bus volume for that date in the available dataset.

Interpretation:
This decision preserved the full traffic history while adding transit-related context where available. It also avoided dropping traffic records simply because a supporting dataset did not have matching records. This is important because the project's core business questions are centered on facility usage, violations, congestion, and forecasting, so traffic records needed to remain complete.

Speed Dataset Review and Integration Decision

The Speeds dataset was reviewed because it provides direct mobility-performance information. It contains monthly facility-level speed measures, including average speed. In principle, this dataset is useful for congestion analysis because lower speeds may indicate congestion pressure at specific facilities.

However, during integration, the speed merge did not successfully produce matched values in the master dataset. The attempted merge resulted in **0 matched average speed records**, meaning the Avg_Speed column became fully missing in the integrated master table. Because the column had no non-null values after merging, it could not be imputed or used reliably inside the final master dataset.

This issue was documented instead of hidden. The likely cause was a key-alignment problem between the master dataset and the Speeds dataset, such as differences in facility identifiers, facility aggregation, or monthly keys. Because the issue affected all matched speed values, using the merged speed column would have created a false sense of completeness.

Therefore, the project treated integrated Avg_Speed as unusable in the current master table. The Speeds dataset was kept as a separate supporting dataset for descriptive mobility interpretation, but it was not used as a populated feature in the final integrated master dataset.

Interpretation:
This was the correct decision. A fully missing merged column should not be filled artificially, especially when there are no valid matched values from which to calculate a reliable replacement. Preserving this issue as a documented limitation protects the integrity of the analysis. It also shows that the project did not force a dataset into the model when the join logic did not produce valid evidence.

Final Master Dataset Validation

After Traffic, Weather, Passenger, and Bus data were aligned and integrated, the final master dataset was validated for row integrity, missing values, duplicates, and string-field cleanliness.

The final dataset maintained the Facility × Date structure and contained **35,168 rows**. Duplicate row checks returned **zero duplicate rows**, confirming that the integration process did not inflate the dataset or create repeated facility-date records.

The final validation also checked object and string fields to make sure there were no hidden blank values or whitespace-only values. The object/string columns detected were STATION and Month_Year. Both fields passed the check, with no blank, whitespace-only, or missing string values.

The final dataset included core traffic variables, weather variables, passenger volume, bus volume, and time-alignment fields. Average speed was excluded from the usable integrated dataset because the speed merge produced no valid matched values.

Interpretation:
The final validation confirmed that the integrated dataset was structurally clean and ready for the next phase. The master dataset preserved traffic completeness, added weather and transit context, avoided duplicate inflation, and documented the speed integration limitation clearly.

Final Integrated Dataset Structure

The final master dataset was designed to support modeling, sampling, dashboard preparation, and business-question analysis. It included the following major groups of variables:

Variable Group	Included Information	Purpose
Facility and Date Fields	Facility code, date, month-year	Supports facility-level and time-based analysis
Traffic Measures	Total traffic, cash, E-ZPass, violations	Core facility usage and toll behavior analysis
Vehicle Categories	Autos, small trucks, large trucks, buses	Supports vehicle mix and usage driver analysis
Weather Variables	Wind, precipitation, snow, max temperature, min temperature, average temperature	Adds external daily conditions
Transit Context	Passenger volume and bus volume	Adds supporting mobility and commuter-demand context
Validation Fields	Cleaned date/month fields and standardized structure	Supports reliable merging and modeling

This structure created a single analytical table where every row had a consistent interpretation: one facility on one date.

Data Cleaning and Integration Conclusion

The data cleaning and integration phase converted several structurally different datasets into one coherent analytical foundation. The most important decision was defining the final grain as Facility × Date. This decision controlled the entire cleaning process and prevented incorrect joins, duplicated rows, inflated totals, and distorted metrics.

The Traffic dataset was preserved as the backbone because it directly answered the core Port Authority questions. It was aggregated from lane-time level to daily facility-level totals, and total traffic volume was fully preserved. Weather was added at the daily level because it aligned naturally with the selected grain and provided external explanatory context. Passenger and Bus datasets were transformed from weekly carrier-level records into daily representations and merged as supporting context. The Speeds dataset was evaluated, but because average speed did not successfully match into the master dataset, it was not forced into the final integrated table.

The final result was a clean, validated, integrated Facility × Date dataset with **35,168 rows**, zero duplicate records, preserved traffic totals, aligned weather and transit context, and documented limitations. This dataset was ready for the next phase of the project: sampling, modeling, forecasting, dashboard development, and final answers to Port Authority's business questions.

Sampling Strategy and Sample Validation

After the data cleaning, grain alignment, and integration phase, the project produced a clean master dataset at the **Facility × Date** level. Each row represented one Port Authority facility on one specific date, with traffic, payment, vehicle-category, weather, passenger-volume, and bus-volume information aligned into one analytical structure.

The final master dataset contained **35,168 rows and 20 columns**. This dataset was complete enough for modeling, but using the full dataset for repeated model experiments, diagnostics, feature testing, and AutoML runs would be less efficient. Sampling was therefore introduced as a controlled step to support faster experimentation while preserving the same operational patterns found in the full dataset.

The purpose of sampling was not to reduce the quality of the analysis. The purpose was to create a smaller, representative version of the full dataset that could be used for rapid modeling iteration while still preserving the structure of the original data. Because Port Authority's questions depend on facility patterns, seasonality, traffic volume, violations, vehicle mix, and external conditions, the sample had to be validated carefully before being used for modeling.

Why Sampling Was Needed

Sampling was used because the cleaned master dataset was large enough that repeated modeling experiments could become slower and less efficient. The project required multiple downstream steps, including feature engineering, model testing, AutoML experimentation, performance comparison, and model interpretation. Working with a representative sample made those steps faster while still keeping the analytical story consistent with the full dataset.

However, sampling can introduce risk if it is done incorrectly. A simple random sample could accidentally overrepresent some facilities, underrepresent certain months, miss seasonal patterns, or distort important traffic and violation distributions. Because this project depends heavily on facility-level and time-based behavior, the sampling method had to preserve both dimensions.

For that reason, the sampling design was based on two principles:

1. **Facility representation must be preserved.**
   Each Port Authority facility needed to remain properly represented in the sample so that the model would not become biased toward only the highest-volume facilities.

2. **Time and seasonality must be preserved.**
   Traffic patterns vary across months and years, so the sample needed to preserve monthly coverage across the full date range.

These requirements led to the use of a stratified sampling approach.

Sampling Method Selected

The project used **stratified sampling** based on:

Facility × Month

This means the full dataset was divided into groups based on facility and month, and a proportional sample was taken from each group. This approach was selected instead of simple random sampling because it protects the two most important structural dimensions in the dataset: where traffic occurs and when traffic occurs.

The sample was drawn at approximately **10%** of the full dataset within each Facility × Month group. The final sampled dataset contained **3,471 rows and 21 columns**.

Dataset	Rows	Columns	Purpose
Full Master Dataset	35,168	20	Complete cleaned Facility × Date dataset
Sample Dataset	3,471	21	Representative modeling sample

The sample also included a DayOfWeek field to support later validation of weekday and weekend balance.

Interpretation:
The sampling method was appropriate because it protected both facility coverage and monthly time coverage. This was important for Port Authority because facility-level differences and seasonal patterns are central to the project. A simple random sample

may have been easier, but it would not have provided the same protection against facility or time-period imbalance.

FAC	DATE	TOTAL	CASH	EZPASS	VIOLATIO	Autos	Small_T	Large_T	Buses	STATION	AWND	PRCP	SNOW	TMAX	TMIN	TAVG	Passenger	Month_Ye	Bus_Volu	DayOfWeek
1	4/24/2025	42466	9671	32071	724	40645	1594	44	183	USW00094	1.4	0	0	25	13.3	19.15	104813	4/1/2025	2974	Thursday
2	4/24/2025	53572	7563	45202	807	44189	3275	504	5604	USW00094	1.4	0	0	25	13.3	19.15	104813	4/1/2025	2974	Thursday
4	4/24/2025	11718	2091	9445	182	10753	477	398	90	USW00094	1.4	0	0	25	13.3	19.15	104813	4/1/2025	2974	Thursday
5	4/24/2025	52455	7451	44025	979	45757	3012	3276	410	USW00094	1.4	0	0	25	13.3	19.15	104813	4/1/2025	2974	Thursday
6	4/24/2025	41415	4310	36602	503	38920	1164	1268	63	USW00094	1.4	0	0	25	13.3	19.15	104813	4/1/2025	2974	Thursday
7	4/24/2025	82005	15294	64926	1785	65295	6360	9779	571	USW00094	1.4	0	0	25	13.3	19.15	104813	4/1/2025	2974	Thursday
9	4/24/2025	59428	9398	48743	1287	58930	142	17	339	USW00094	1.4	0	0	25	13.3	19.15	104813	4/1/2025	2974	Thursday
1	5/28/2025	40399	8178	28817	3404	38415	1752	16	216	USW00094	1.4	12.4	0	18.3	11.1	14.7	104671	5/1/2025	2985	Wednesday
2	5/28/2025	52445	7542	41538	3365	42886	3085	771	5703	USW00094	1.4	12.4	0	18.3	11.1	14.7	104671	5/1/2025	2985	Wednesday
4	5/28/2025	11385	1777	8769	839	10488	469	334	94	USW00094	1.4	12.4	0	18.3	11.1	14.7	104671	5/1/2025	2985	Wednesday
5	5/28/2025	50213	7091	39632	3490	43981	3102	2693	437	USW00094	1.4	12.4	0	18.3	11.1	14.7	104671	5/1/2025	2985	Wednesday
6	5/28/2025	40032	4029	33459	2544	37647	1147	1159	79	USW00094	1.4	12.4	0	18.3	11.1	14.7	104671	5/1/2025	2985	Wednesday
7	5/28/2025	79850	14695	59240	5915	63062	6673	9567	548	USW00094	1.4	12.4	0	18.3	11.1	14.7	104671	5/1/2025	2985	Wednesday
9	5/28/2025	56061	8800	42160	5101	55545	181	11	324	USW00094	1.4	12.4	0	18.3	11.1	14.7	104671	5/1/2025	2985	Wednesday
1	9/24/2024	41846	8615	31482	1749	40037	1613	20	176	USW00094	2.5	0	0	20	15	17.5	104177	9/1/2024	2938	Tuesday
2	9/24/2024	53781	7631	44272	1878	44154	3559	641	5427	USW00094	2.5	0	0	20	15	17.5	104177	9/1/2024	2938	Tuesday
4	9/24/2024	11593	1737	9405	451	10608	531	378	76	USW00094	2.5	0	0	20	15	17.5	104177	9/1/2024	2938	Tuesday
5	9/24/2024	47954	6423	39779	1752	41321	2925	3341	367	USW00094	2.5	0	0	20	15	17.5	104177	9/1/2024	2938	Tuesday
6	9/24/2024	40364	3830	35468	1066	37731	1279	1303	51	USW00094	2.5	0	0	20	15	17.5	104177	9/1/2024	2938	Tuesday
7	9/24/2024	68528	12490	53489	2549	52881	6463	8653	531	USW00094	2.5	0	0	20	15	17.5	104177	9/1/2024	2938	Tuesday
9	9/24/2024	67020	13330	50859	2831	66543	73	18	386	USW00094	2.5	0	0	20	15	17.5	104177	9/1/2024	2938	Tuesday
1	10/28/2024	41823	8473	31483	1867	40075	1579	21	148	USW00094	1.3	0	0	16.1	7.8	11.95	103931	########	2948	Monday
2	10/28/2024	53107	7463	43804	1840	43848	3230	572	5457	USW00094	1.3	0	0	16.1	7.8	11.95	103931	########	2948	Monday
4	10/28/2024	11140	1659	9014	467	10191	478	401	70	USW00094	1.3	0	0	16.1	7.8	11.95	103931	########	2948	Monday
5	10/28/2024	50205	7040	41261	1904	43856	2847	3123	379	USW00094	1.3	0	0	16.1	7.8	11.95	103931	########	2948	Monday
6	10/28/2024	39253	3694	34399	1160	36975	1091	1145	42	USW00094	1.3	0	0	16.1	7.8	11.95	103931	########	2948	Monday
7	10/28/2024	68962	12506	53717	2739	53753	6146	8545	518	USW00094	1.3	0	0	16.1	7.8	11.95	103931	########	2948	Monday
9	10/28/2024	63538	11797	48824	2917	63122	48	12	356	USW00094	1.3	0	0	16.1	7.8	11.95	103931	########	2948	Monday
1	4/28/2025	40063	7941	30996	1126	38318	1537	27	181	USW00094	1.5	0	0	24.4	10.6	17.5	103472	4/1/2025	2962	Monday
2	4/28/2025	49653	7053	41590	1010	40479	3269	494	5411	USW00094	1.5	0	0	24.4	10.6	17.5	103472	4/1/2025	2962	Monday
4	4/28/2025	11181	2052	8911	218	10272	460	366	83	USW00094	1.5	0	0	24.4	10.6	17.5	103472	4/1/2025	2962	Monday

Representativeness Validation Approach

After drawing the sample, the next step was to prove that the sample still represented the full dataset. This validation was necessary because the sample would later be used for modeling and AutoML experimentation. If the sample did not match the full dataset, model results could become misleading.

The sample was validated in four ways:

Validation Area	Purpose
Numeric summary comparison	Check whether key means, medians, standard deviations, minimums, and maximums stayed close to the full dataset
Facility distribution comparison	Confirm that each facility remained represented at nearly the same proportion
Time distribution comparison	Confirm that monthly coverage and seasonality were preserved
Statistical distribution testing	Use KS tests to compare full dataset and sample distributions for numeric variables
Visual distribution comparison	Use density plots and bar charts to visually confirm overlap between full and sampled data

This multi-step validation was important because no single test is enough to confirm a sample's quality. Summary statistics show whether averages and medians are close. Facility and month distributions show whether the structure is preserved. Statistical tests show whether distributions are meaningfully different. Visual charts help confirm whether the sample follows the same overall shape as the full dataset.

Numeric Summary Comparison

The numeric comparison showed that the sample closely matched the full dataset across the most important traffic and volume variables.

For total traffic, the full dataset had an average of approximately **41,357 vehicles per facility-date**, while the sample had an average of approximately **41,382 vehicles**. The difference was only approximately **0.06%**, which is extremely small. This confirms that the sample preserved the central tendency of the main usage variable.

Variable	Full Dataset Mean	Sample Mean	Difference
Total Traffic	41,357.21	41,382.04	+0.06%
Autos	37,772.98	37,841.83	+0.18%
Cash	5,566.33	5,595.48	+0.52%
Violations	2,306.23	2,311.45	+0.23%
Buses	893.07	885.19	-0.88%
Small Trucks	1,377.29	1,359.90	-1.26%
Large Trucks	1,313.86	1,295.12	-1.43%
Passenger Volume	17,690.62	17,273.88	-2.36%
Bus Volume	642.88	629.03	-2.15%

The most important traffic variables showed very small differences between the full dataset and the sample. Total traffic, autos, cash, violations, and vehicle categories were all very closely aligned. Passenger and bus volume also remained close, with differences of approximately **-2.36%** and **-2.15%**, respectively.

Weather variables showed slightly larger differences for some measures, especially snow, precipitation, and average temperature. However, the absolute structure remained reasonable, and the core traffic variables were strongly preserved.

Facility Representation

Facility representation was one of the most important validation checks because Port Authority's facilities do not all behave the same way. Some facilities carry much higher traffic volumes, some have higher violation rates, and others show different payment behavior. If the sample overrepresented or underrepresented a facility, the model could become biased.

The facility distribution comparison showed that the sample matched the full dataset almost perfectly. The maximum deviation across facilities was approximately **0.09 percentage points**, and most facility differences were around **0.01 percentage points**.

Interpretation:
This confirms that the stratified sampling strategy successfully preserved facility-level representation. The sample did not accidentally become biased toward one facility, one bridge group, one tunnel group, or one George Washington Bridge segment. This is critical because facility-level variation is central to the project.

FAC distribution comparison:

FAC	full_pct	sample_pct	diff_pct_points
1	0.128924	0.128781	-0.014269
2	0.128924	0.128781	-0.014269
4	0.127588	0.128493	0.090565
5	0.128924	0.128781	-0.014269
6	0.128924	0.128781	-0.014269
7	0.128924	0.128781	-0.014269
8	0.098868	0.098819	-0.004951
9	0.128924	0.128781	-0.014269

Full Dataset vs Sample Dataset Facility Distribution Comparison

Monthly and Seasonal Representation

The second major validation check was monthly representation. This was important because traffic behavior changes by season. Earlier analysis showed that traffic tends to rise through spring, peak during late spring and summer, and decline during late fall and winter. Therefore, the sample needed to preserve month-level coverage.

The month-by-month comparison showed that the sample preserved monthly representation extremely well. For the first twelve months shown in the validation output, most deviations were approximately **0.01 percentage points**, with the largest shown deviation around **0.05 percentage points**.

Interpretation:
This confirms that the sample preserved the long-term time structure of the full dataset. Since the sample was stratified by Facility × Month, it retained both facility coverage and seasonal coverage. This means the sample is suitable for modeling patterns that depend on monthly seasonality and long-term changes over time.

Month_Year	full_pct	sample_pct	diff_pct_points
2013-01-01	0.007052	0.006914	-0.013743
2013-02-01	0.006369	0.006914	0.054501
2013-03-01	0.007052	0.006914	-0.013743
2013-04-01	0.006824	0.006914	0.009005
2013-05-01	0.007052	0.006914	-0.013743
2013-06-01	0.006824	0.006914	0.009005
2013-07-01	0.007052	0.006914	-0.013743
2013-08-01	0.007052	0.006914	-0.013743
2013-09-01	0.006824	0.006914	0.009005
2013-10-01	0.007052	0.006914	-0.013743
2013-11-01	0.006824	0.006914	0.009005
2013-12-01	0.007052	0.006914	-0.013743

Full Dataset vs Sample Dataset Month_Year Distribution Comparison

---

Day-of-Week Balance

Although the sample was stratified by Facility × Month, it was also checked for day-of-week balance. This was important because traffic can differ between weekdays and weekends. Earlier analysis showed that traffic tends to peak around Friday and remains strong on weekends, while E-ZPass usage tends to decline on weekends.

The day-of-week distribution remained close overall. The validation found a small shift in a few days, especially:

Day	Sample Drift
Thursday	+0.58 percentage points
Friday	-0.51 percentage points

Other days showed smaller differences.

Interpretation:
This small drift does not represent a sampling failure. It happened because the sample was stratified by Facility × Month, not by DayOfWeek. The selected strategy prioritized preserving facility and seasonal structure, which were more important for the main project questions and modeling setup. The day-of-week differences were small enough that the sample remained reliable. However, for highly detailed weekday-versus-weekend analysis, the full dataset may still be preferred.

```
        DayOfWeek distribution comparison:
                      full_pct    sample_pct  diff_pct_points
        DayOfWeek
Friday        0.143113    0.138001        -0.511248
Monday        0.142886    0.142034        -0.085158
Saturday      0.142345    0.143186         0.084109
Sunday        0.142317    0.143763         0.144573
Thursday      0.143113    0.148948         0.583537
Tuesday       0.143113    0.143475         0.036145
Wednesday     0.143113    0.140593        -0.251957
```
Full Dataset vs Sample Dataset Day-of-Week Distribution Comparison

---

Statistical Distribution Validation: KS Test

To formally compare numeric distributions, the project used Kolmogorov-Smirnov tests, commonly referred to as KS tests. The purpose was to check whether the sample distribution differed meaningfully from the full dataset distribution for each numeric variable.

The most important modeling variables showed very strong alignment. Total traffic had a KS statistic of approximately **0.0102** and a p-value of approximately **0.8969**. This means the total traffic distribution in the sample was very close to the full dataset distribution. Other core variables, including autos, small trucks, large trucks, buses, cash, E-ZPass, and violations, also showed very strong alignment, with generally high p-values and low KS statistics.

Variable	KS Statistic	p-value	Interpretation
Total Traffic	0.0102	0.8969	Strong match
Autos	0.0113	0.8076	Strong match
Buses	0.0097	0.9266	Strong match
Large Trucks	0.0097	0.9270	Strong match
Small Trucks	0.0087	0.9685	Strong match
Violations	0.0084	0.9769	Strong match
E-ZPass	0.0080	0.9860	Strong match
Cash	0.0076	0.9931	Strong match
Passenger Volume	0.0125	0.7051	Strong match
Bus Volume	0.0118	0.7632	Strong match

A few weather variables showed statistically detectable differences:

Variable	KS Statistic	p-value	Interpretation
AWND	0.0293	0.0088	Small but statistically detectable drift
TAVG	0.0254	0.0332	Small but statistically detectable drift
PRCP	0.0246	0.0425	Small but statistically detectable drift

These differences were noted, but the KS statistics were still small, generally around **0.02 to 0.03**. With large datasets, statistical tests can flag very small differences as significant even when the practical difference is limited. Therefore, these results indicate minor weather distribution drift, not a failure of the sampling process.

Interpretation:
The KS results strongly support the quality of the sample for traffic modeling. The core traffic and volume variables match the full dataset very closely. Weather variables show minor drift, but the magnitude is small and does not undermine the overall representativeness of the sample.

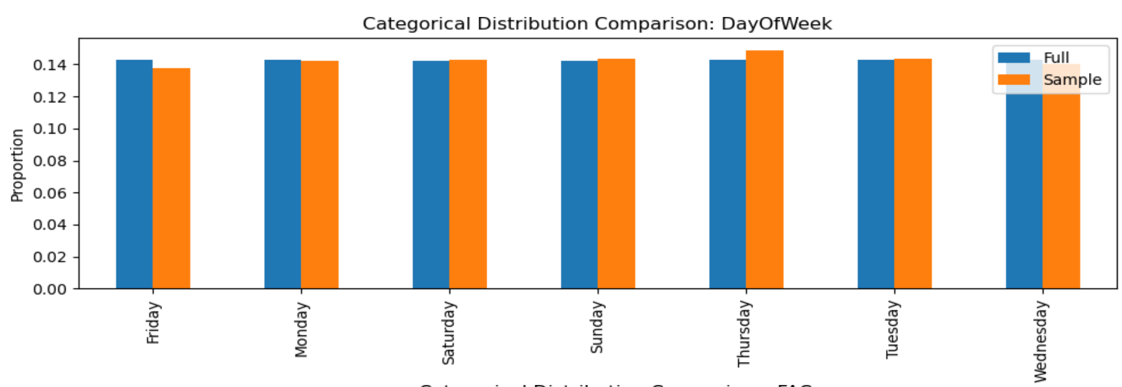
Visual Distribution Validation

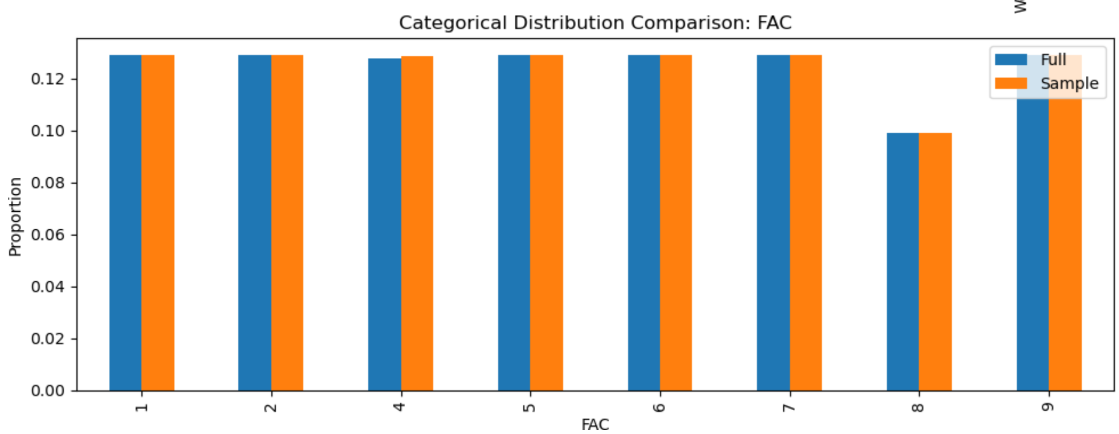
In addition to statistical validation, visual distribution comparisons were created for the full dataset and the sample dataset. These charts used density-based distributions rather than raw counts because the full dataset contained far more rows than the sample. Density plots allowed the shape of each distribution to be compared fairly.

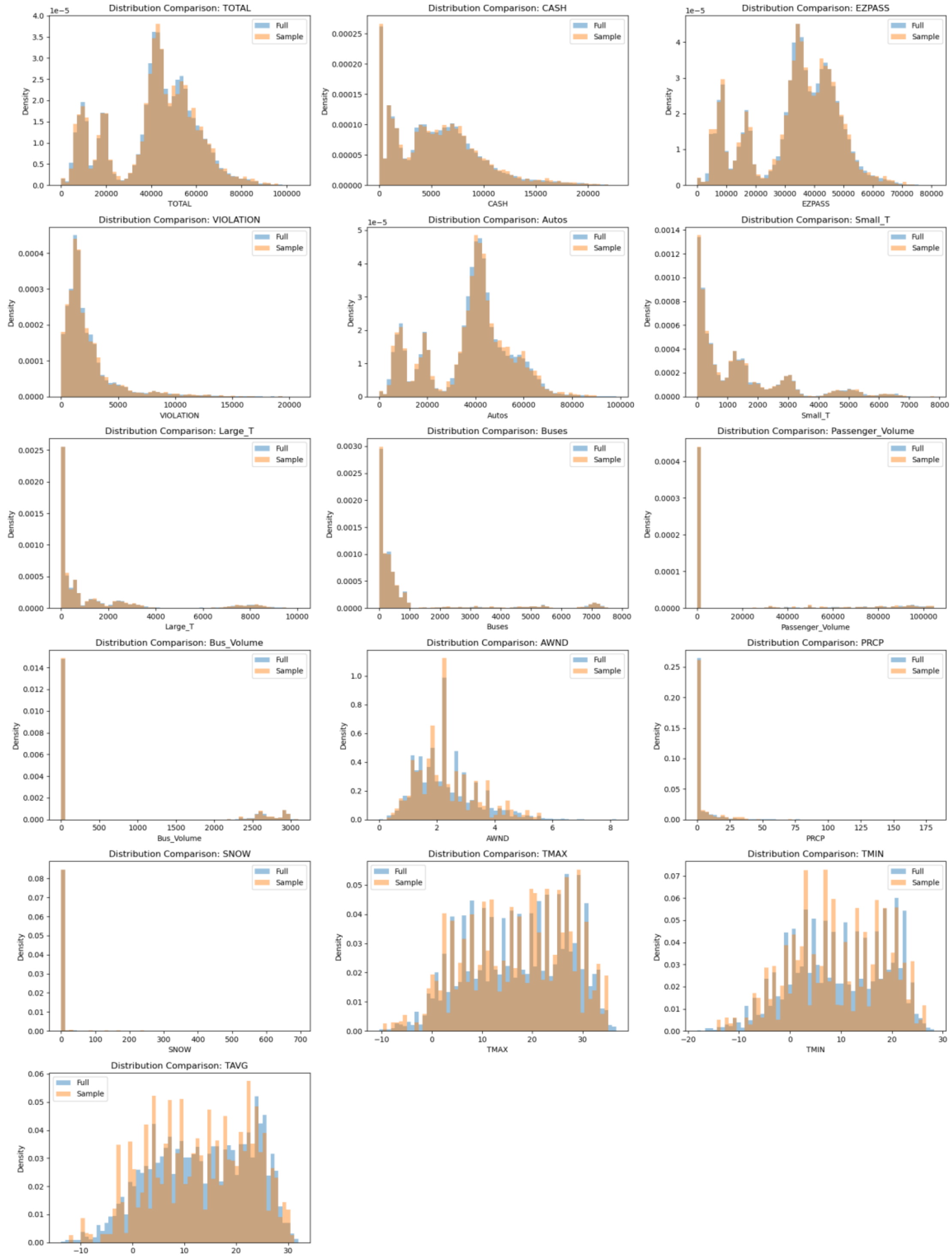
The distribution charts showed strong overlap between the full dataset and the sample across the primary traffic, vehicle-category, payment, passenger-volume, bus-volume, and external-driver variables. This visual overlap confirmed that the sample preserved the same overall distributional patterns rather than distorting the data.

Categorical bar charts were also used for facility and day-of-week distributions. These charts visually confirmed that the sample preserved facility representation extremely well and maintained day-of-week balance with only minor drift.

Interpretation:
The visual validation supports the statistical results. The sample is not only numerically close to the full dataset; it also visually follows the same distributional shape. This is important because models are sensitive not only to averages but also to spread, skewness, and distribution shape.







Density Comparison Charts: Full Dataset vs Sample Dataset

Final Sampling Decision

Based on the validation results, the sample was considered representative and suitable for downstream modeling work.

The sample successfully preserved:

- Facility coverage
- Monthly and seasonal coverage
- Core traffic volume distributions
- Payment-channel distributions
- Vehicle-category distributions
- Passenger and bus volume patterns
- Overall distributional shape

The sample showed minor drift in day-of-week proportions and small statistically detectable differences in a few weather variables. However, these differences were limited and did not undermine the sample's overall representativeness. The most important project variables, especially total traffic, autos, cash, E-ZPass, violations, passenger volume, and bus volume, showed strong alignment between the full dataset and the sample.

The final sampled dataset contained **3,471 rows and 21 columns** and was saved as the modeling sample for the next phase.

Sampling Conclusion

Sampling was completed in a controlled and statistically defensible way. The project did not use a random shortcut. Instead, it used stratified sampling by Facility × Month to protect the two most important structural dimensions in the data: facility behavior and time-based seasonality.

The validation results confirmed that the sample remained highly representative of the full cleaned master dataset. Facility representation was preserved almost perfectly. Monthly coverage was preserved extremely well. Core numeric traffic and volume variables matched closely. Statistical testing and visual distribution checks both supported the sample's quality.

Therefore, the sampled dataset was appropriate for rapid modeling iteration, AutoML experiments, feature engineering, model diagnostics, and early model selection. The full dataset remains the complete analytical source, but the sample provides a reliable and efficient modeling environment for the next stage of the project.

Model Development : Final Answers to Project Questions.

After the data audit, exploratory analysis, data cleaning, integration, and sampling phases, the project moved into the model development and final answer phase. The purpose of this phase was to answer Port Authority's business questions using the most appropriate analytical method for each question.

Not every project question required the same type of model. Some questions asked for actual counts and patterns, which are best answered through descriptive analytics. Other questions required prediction or classification, where machine learning was appropriate. Forecasting future usage required a time-series approach. Therefore, the method was selected based on the business purpose of each question, not based on using one model for every problem.

The validated sample dataset contained **3,471 rows and 21 columns** and was used for modeling because it preserved facility coverage, monthly time coverage, seasonal structure, and the major traffic-volume distributions from the full dataset. The dataset included facility, date, total traffic, payment channels, vehicle categories, weather variables, passenger volume, bus volume, month-year, and day-of-week fields. No additional sampling was performed during model development.

The facility codes used in the modeling output were translated into facility names for final reporting:

Facility Code	Facility Name
1	Holland Tunnel
2	Lincoln Tunnel
4	Bayonne Bridge
5	Goethals Bridge
6	Outerbridge Crossing
7	George Washington Bridge Upper
8	George Washington Bridge PIP

9	George Washington Bridge Lower

Project Question 1

What are the top five factors that affect the usage of bridges and tunnels by drivers?

Business Purpose

Port Authority wanted to understand which factors most strongly influence bridge and tunnel usage. This is important because usage patterns affect congestion planning, staffing, toll operations, maintenance scheduling, and long-term demand planning.

Because the target variable, **TOTAL traffic**, is continuous, this question was treated as a regression problem. The goal was to identify which variables are most useful for explaining or predicting facility usage.

Modeling Approach

H2O AutoML was used to test multiple regression models and compare their performance objectively. The goal was not to manually select a model based on preference, but to allow AutoML to evaluate different model families and identify the strongest predictive approach.

A key modeling decision was made before running the model: direct components of total traffic were excluded from the predictor list. Variables such as **CASH, E-ZPass, VIOLATION, Autos, Small_T, Large_T, and Buses** are direct parts of total traffic. Including them would allow the model to mechanically reconstruct TOTAL instead of identifying meaningful external drivers. For that reason, the model used facility, day-of-week, weather variables, passenger volume, and bus volume as predictors. This made the model more useful for business interpretation.

The model used the following predictors:

Predictor Group	Variables Used	Why Included
Facility	FAC	Captures location-based demand differences
Time Pattern	DayOfWeek	Captures weekday/weekend and travel-behavior patterns

Weather	AWND, PRCP, SNOW, TMAX, TMIN, TAVG	Captures external environmental conditions
Transit Context	Passenger_Volume, Bus_Volume	Captures supporting movement patterns

The best-performing model from AutoML was a **Stacked Ensemble**, while the best interpretable non-ensemble model was a **Gradient Boosting Machine (GBM)**. The Stacked Ensemble was strongest for prediction, but the GBM was used for variable importance because it provides clearer interpretability. The AutoML leaderboard showed the Stacked Ensemble with the lowest RMSE, while the GBM was very close behind and more interpretable for management explanation.

Model Performance

The Stacked Ensemble model achieved strong predictive performance, with cross-validation R^2 of approximately **0.944**, showing that the selected predictors explained a large share of traffic variation. The best interpretable GBM model had a test RMSE of approximately **4,001 vehicles** and MAE of approximately **2,823 vehicles**, which is reasonable given that traffic values vary widely across facilities and dates.

Final Answer

The top five factors affecting usage of Port Authority bridges and tunnels were:

Rank	Factor	Interpretation
1	Facility	The strongest driver of usage; traffic varies heavily by crossing location
2	Day of Week	Usage changes based on weekday, weekend, commuter, and leisure patterns
3	Maximum Temperature	Warmer or colder conditions align with seasonal travel behavior
4	Average Temperature	Reinforces seasonal movement patterns
5	Wind Speed	Weather-related operating condition, especially relevant for exposed bridge facilities

The most important finding is that **facility location is the dominant driver of usage**. This means traffic demand is not evenly distributed across Port Authority crossings. Some facilities naturally carry more volume because of their geography, connectivity, commuter role, and route importance.

Day of week was the second most important factor, confirming that usage is shaped by recurring travel behavior. Weather variables also contributed, but they were less influential than facility and time. This means weather matters, but it does not replace structural facility demand as the main driver.

Recommendation for Port Authority

Port Authority should use a **facility-first demand monitoring approach**. This means usage planning should begin at the facility level rather than with system-wide averages. The model shows that facility is the strongest driver of traffic, so system-wide averages can hide the facilities that are carrying the greatest operational burden.

To implement this, Port Authority should maintain a recurring facility-level demand dashboard that tracks daily traffic, day-of-week trends, seasonal movement, and weather-sensitive changes by facility. George Washington Bridge Upper, George Washington Bridge Lower, Lincoln Tunnel, and other high-volume facilities should be reviewed separately instead of being blended into one system total.

Operational planning should also include day-of-week patterns. Since demand changes by weekday and weekend, staffing, lane management, maintenance windows, and enforcement scheduling should be adjusted based on expected day-level patterns. For example, routine non-urgent operational activities should be planned around lower-demand periods rather than peak Fridays, Saturdays, or summer travel months.

Weather should be used as a supporting signal. Temperature and wind should not be treated as the primary cause of traffic demand, but they should be included in daily planning dashboards because they can help explain short-term changes, especially during adverse weather periods.

Project Question 2

How many toll violators are there by year, month, week, and facility?

Business Purpose

Port Authority wanted to understand the scale and distribution of toll violations. This question asked for actual violation counts by time interval and facility, so the appropriate method was descriptive analytics, not predictive modeling.

Using a model here would not add value because the question is not asking what future violations will be or which class a record belongs to. It is asking **how many violations occurred, where they occurred, and when they occurred**.

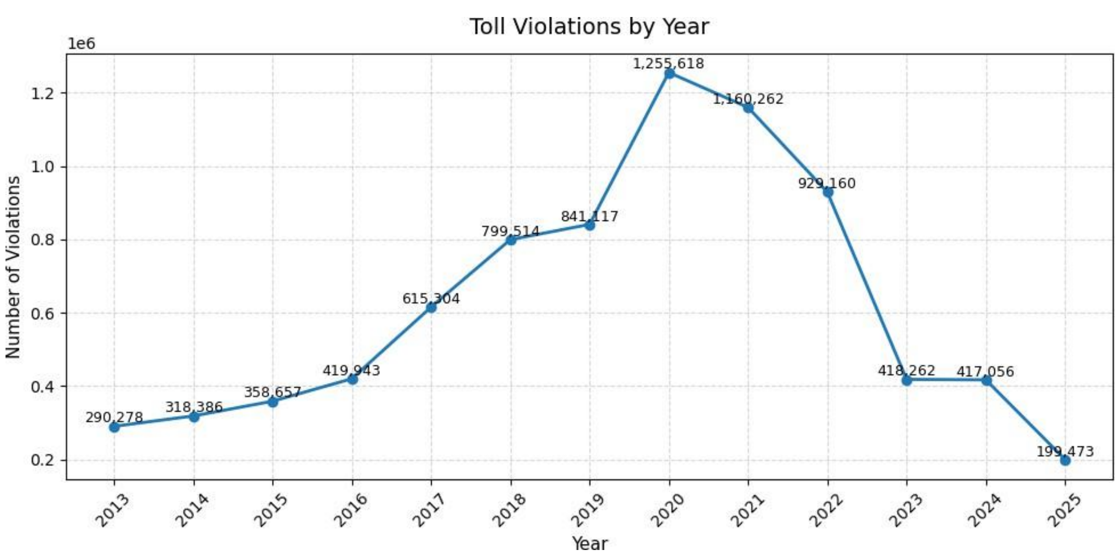
Analytical Approach

Violation counts were aggregated by:

- Year
- Month
- Week
- Facility

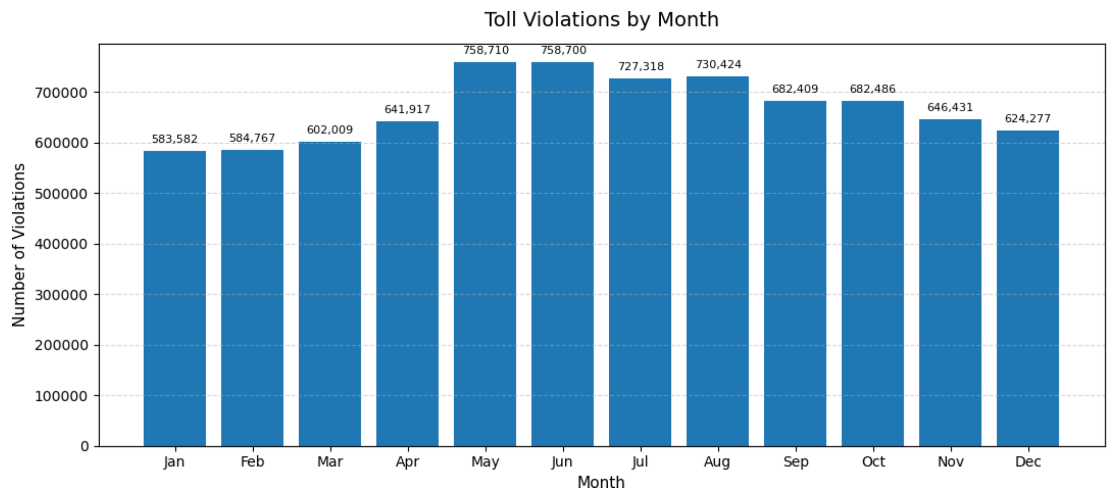
Facility-level contribution percentages were calculated to identify where violations were concentrated. Visualizations were used to show yearly trends, seasonal patterns, weekly patterns, and facility-level concentration.

Final Answer



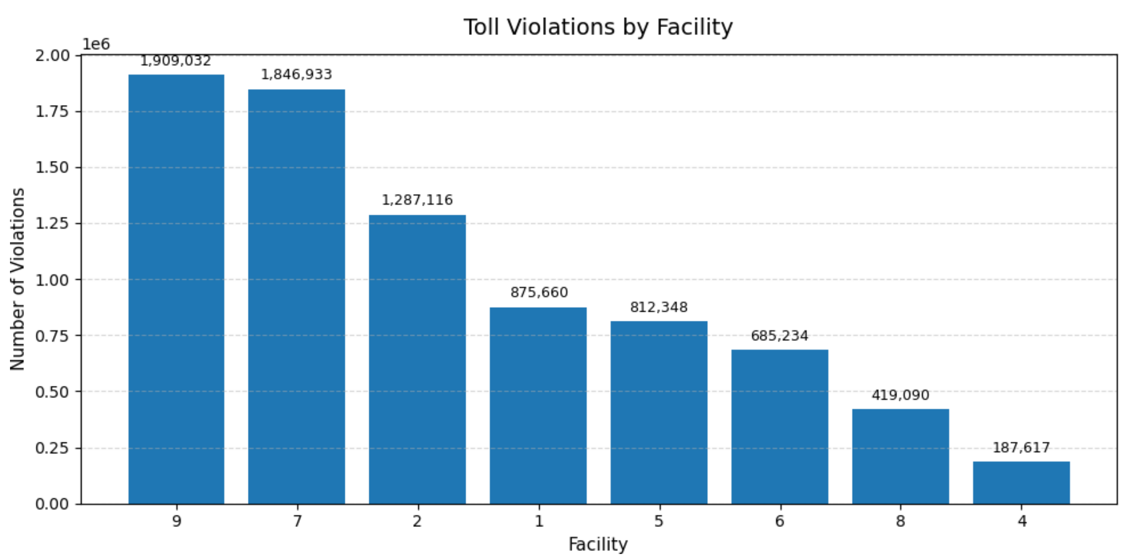
The analysis identified **8,023,030 total toll violations** in the analyzed dataset. This confirms that toll violations are not isolated incidents; they represent a large operational and revenue-protection issue.

The yearly trend showed that violations increased steadily from approximately **290,000 in 2013** to approximately **1.25 million around 2020**, followed by a decline after 2021. This suggests that violations followed a long-term growth pattern before the disruption period, with 2020 acting as a major behavioral or operational turning point.



Monthly analysis showed that violations were higher during **May through August** and lower during **January and February**. This suggests that violations rise during higher-travel months, which is consistent with seasonal demand patterns.

Facility-level analysis showed that violations were highly concentrated:



Facility	Share of Total Violations	Interpretation
George Washington Bridge Lower	23.79%	Highest violation concentration
George Washington Bridge Upper	23.02%	Second-highest violation concentration
Lincoln Tunnel	16.04%	Major violation contributor
Holland Tunnel	10.91%	Meaningful violation contributor
Goethals Bridge	10.13%	Meaningful violation contributor
Outerbridge Crossing	8.54%	Moderate contribution
George Washington Bridge PIP	5.22%	Lower contribution
Bayonne Bridge	2.34%	Lowest contribution

The most important finding is that **George Washington Bridge Lower and George Washington Bridge Upper together account for nearly 47% of total violations**. This means violation behavior is highly concentrated rather than evenly distributed across the system.

Recommendation for Port Authority

Port Authority should prioritize toll-violation monitoring at the facilities where violations are concentrated, especially **George Washington Bridge Lower, George Washington Bridge Upper, and Lincoln Tunnel**.

This should be done in three steps.

First, create a facility-level violation monitoring dashboard that updates by year, month, week, and day. The dashboard should show total violations, violation share, violation rate, and change from prior periods. This allows management to see whether violations are increasing due to higher traffic volume or due to worsening compliance behavior.

Second, focus enforcement review on the highest-concentration facilities. Since George Washington Bridge Lower and George Washington Bridge Upper together represent almost half of total violations, even a small improvement at those facilities would likely have a larger system impact than spreading attention evenly across all crossings.

Third, align violation monitoring with seasonal peaks. Since violations increase during higher-travel months, monitoring should intensify before and during May through August. This does not necessarily mean only increasing staffing; it can include better signage, improved communication around toll rules, plate image review quality checks, targeted customer reminders, and monitoring of repeat violation patterns.

The key point is that Port Authority should not treat violations as a uniform system-wide issue. The data shows that violations are concentrated by facility and season, so the response should also be targeted by facility and season.

Project Question 3

What are the busiest times throughout the year, and how is traffic affected by seasonality, toll violators, vehicle type, holidays, weather, and other factors?

Business Purpose

Port Authority wanted to understand when traffic reaches its highest levels and what factors are associated with those busy periods. This question is about identifying patterns rather than predicting a future value, so descriptive analytics, aggregation, correlation analysis, and visualization were used.

Analytical Approach

Traffic was analyzed by:

- Year
- Month
- Day of week
- Weekday vs. weekend
- Busiest individual days
- Relationship with violations
- Vehicle composition
- Weather conditions

Final Answer

Year	Avg Traffic
2013	11245850
2014	11323991
2015	11464737
2016	11680453
2017	11733732
2018	11927417
2019	12208156
2020	9633298
2021	11592347
2022	12024689
2023	12183377
2024	11816208
2025	4802818

DayOfWeek	Avg Traffic
Friday	42485.630480
Saturday	42347.004024
Sunday	41767.769539
Wednesday	41553.004098
Thursday	41188.029014
Monday	40561.288032
Tuesday	39817.441767

Month	Avg Traffic
1	37287.669967
2	37061.450331
3	39831.785479
4	40951.386139
5	41946.541254
6	43860.315603
7	44126.007143
8	44455.261649
9	42342.605735
10	42458.340502
11	41691.584229
12	41365.315412

Traffic increased steadily from 2013 through 2019, rising from approximately **11.2 million** to more than **12.2 million** vehicles in the sampled analysis. Traffic dropped sharply in 2020 to approximately **9.6 million**, which represents a major external disruption rather than normal traffic fluctuation. Traffic then recovered in later years, exceeding **12.1 million** again by 2023. The lower 2025 value reflects partial-year data and should not be interpreted as a full-year decline.

Monthly traffic showed a clear seasonal pattern. Traffic was lowest in January and February and increased toward the summer. The highest average monthly traffic occurred between **June and August**, with average daily traffic exceeding approximately **44,000 vehicles** during peak summer months. This confirms that traffic is strongly seasonal.

Day-of-week analysis showed that traffic is relatively stable across the week but slightly higher toward the end of the week. Friday and Saturday had the highest average traffic, while Tuesday had the lowest. Weekend traffic was slightly higher than weekday traffic, indicating that the system is not only commuter-driven. Leisure, discretionary, tourism, family, event, and weekend travel also contribute significantly to traffic demand.

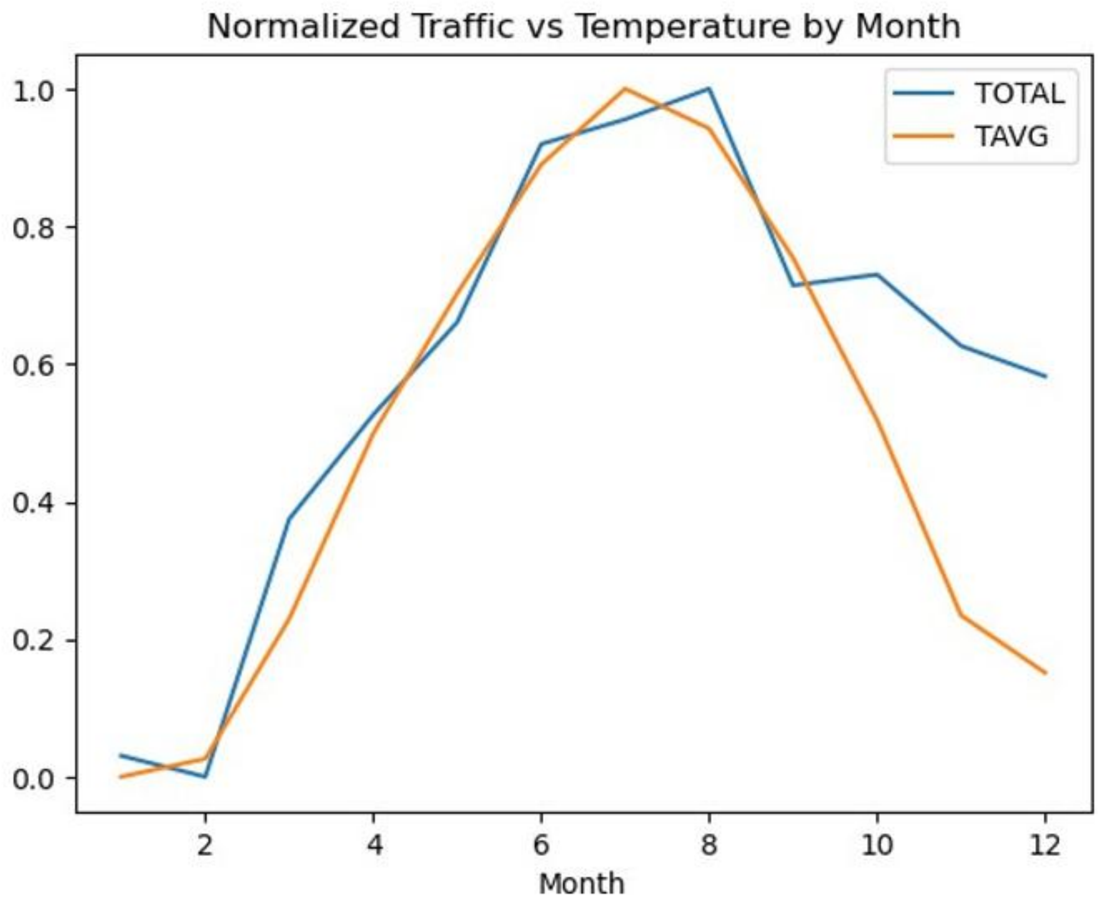
	DATE	TOTAL	FAC	DayOfWeek
	2022-09-24	95214	7	Saturday
	2023-12-16	95182	7	Saturday
	2023-05-28	93844	7	Sunday
	2024-11-16	92482	7	Saturday
	2024-06-16	92479	7	Sunday
	2023-06-24	92408	7	Saturday
	2025-05-24	90519	7	Saturday
	2023-10-28	90250	7	Saturday
	2025-05-16	89825	7	Friday
	2022-11-24	89751	7	Thursday

The top 10 busiest days were concentrated heavily at **George Washington Bridge Upper**, with the maximum observed traffic exceeding **95,000 vehicles** in one day. This confirms that extreme traffic pressure is location-specific and event-driven rather than evenly distributed across all facilities.

The analysis also showed a moderate positive relationship between traffic and violations, with a correlation of approximately **0.48** between total traffic and violation counts in the sample. This suggests that higher traffic volume creates more opportunities for violations and increases enforcement pressure during peak demand periods.

	Autos	Small_T	Large_T	Buses	TOTAL
Autos	1.000000	0.358496	0.318724	0.164755	0.977927
Small_T	0.358496	1.000000	0.858487	0.375090	0.537942
Large_T	0.318724	0.858487	1.000000	-0.031142	0.469668
Buses	0.164755	0.375090	-0.031142	1.000000	0.267658
TOTAL	0.977927	0.537942	0.469668	0.267658	1.000000

Vehicle composition showed that autos are the dominant contributor to traffic variation. Autos had an extremely strong correlation with total traffic, approximately **0.98**, while trucks had moderate relationships and buses had a smaller relationship. This means congestion is primarily driven by passenger vehicle flow rather than commercial or bus traffic.



Weather showed a supporting role. Temperature and traffic followed similar seasonal patterns, increasing together through warmer months and declining toward colder months. However, the statistical relationship between temperature and traffic was relatively weak, around **0.11 to 0.12**, which means weather reinforces seasonal behavior but is not the primary driver of traffic. Snow showed a negative relationship with traffic, indicating reduced travel during adverse weather periods.

Recommendation for Port Authority

Port Authority should plan operations around three recurring demand layers: **seasonality, day-of-week behavior, and facility-specific peak pressure**.

First, summer demand planning should begin before the peak season. Since traffic rises through spring and peaks in June through August, Port Authority should review lane availability, maintenance scheduling, staffing levels, and incident-response readiness before the summer period begins. Non-urgent maintenance that reduces capacity should be scheduled outside the strongest seasonal demand windows whenever possible.

Second, weekend and late-week demand should be treated as operationally important. The analysis shows that weekends are not low-demand periods. Therefore, Friday, Saturday, and Sunday should receive dedicated monitoring in dashboards and staffing plans. This is especially important for facilities with high recreational, event, or discretionary travel exposure.

Third, the George Washington Bridge Upper should receive special peak-day monitoring. Since the busiest individual days were concentrated there, Port Authority should maintain a separate high-demand playbook for that facility. This should include daily peak alerts, pre-positioned incident response during expected high-volume days, and closer monitoring during holiday weekends, summer weekends, and major regional event periods.

Fourth, enforcement and toll-compliance monitoring should be aligned with peak traffic periods. Since violations rise with traffic counts, enforcement review should not be separated from traffic operations. When traffic is expected to rise, violation monitoring should also be reviewed in advance.

Finally, weather should be used as a supporting planning input. Temperature alone should not drive operational decisions, but snow, precipitation, wind, and extreme temperatures should be included in daily readiness reviews because they can affect both traffic flow and safety conditions.

Project Question 4

Report on congestion due to pricing in 2025 by facility. Is there a shift in traffic patterns where drivers try to avoid tolls and congestion happens in other facilities?

Business Purpose

Port Authority wanted to understand whether 2025 traffic patterns shifted across facilities, potentially indicating that drivers changed route choices in response to congestion or pricing conditions. The key question was not simply whether traffic increased or decreased overall, but whether traffic was redistributed across facilities.

Analytical Approach

This question was answered in two parts.

First, facility-level traffic share before 2025 was compared with facility-level traffic share during 2025. This shows whether some facilities gained or lost a larger share of total traffic.

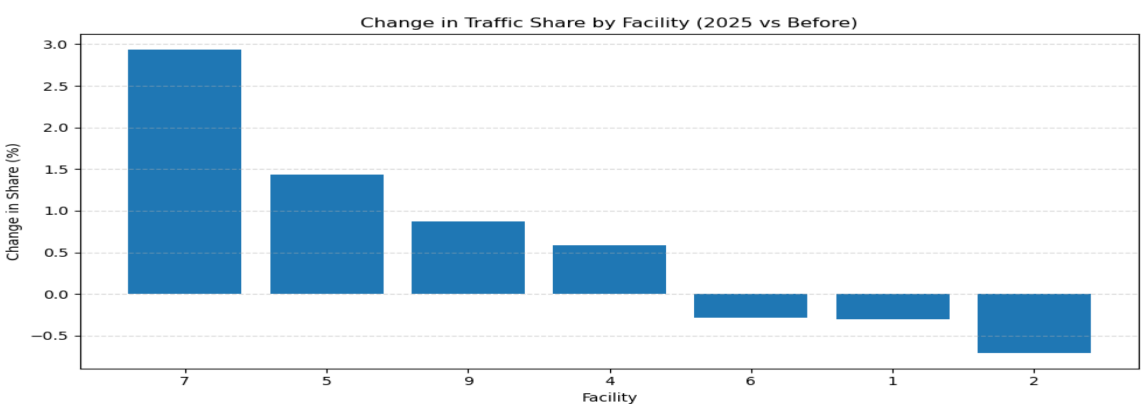
Second, a classification model was developed to identify which factors are associated with high-traffic conditions. High traffic was defined using the **75th percentile of TOTAL traffic**, with values above that threshold treated as congestion or high-traffic conditions. This avoided using an arbitrary congestion cutoff.

The model excluded direct traffic components such as TOTAL, cash, E-ZPass, violations, and vehicle counts to avoid data leakage. The features used included facility, day of week, weather variables, passenger volume, and bus volume.

Final Answer

The analysis showed that traffic distribution changed in 2025 compared with historical patterns.

The largest increases in traffic share were:



Facility	Change in Traffic Share
George Washington Bridge Upper	+2.94 percentage points
Goethals Bridge	+1.43 percentage points
George Washington Bridge Lower	+0.87 percentage points
Bayonne Bridge	+0.59 percentage points

The facilities showing declines included:

Facility	Change in Traffic Share
Lincoln Tunnel	-0.71 percentage points
Holland Tunnel	-0.30 percentage points
Outerbridge Crossing	-0.28 percentage points

These results indicate that traffic did not remain distributed in the same way as previous years. Instead, some facilities gained traffic share while others lost share. This suggests that drivers may be adapting route choices, potentially in response to congestion, pricing, travel conditions, or perceived route efficiency.

The classification model supported this finding. H2O AutoML identified a Stacked Ensemble as the best-performing model for distinguishing high-traffic from normal-traffic conditions. The model achieved AUC of approximately **0.976 to 0.984** and overall accuracy of approximately **93%**, showing that high-traffic conditions were highly predictable from the selected features.

For interpretation, the best non-ensemble model, GBM, was used. The variable importance results showed that **facility accounted for nearly 88% of model importance**, followed by day of week and weather variables. This confirms that congestion is primarily location-driven rather than evenly distributed across the system.

Recommendation for Port Authority

Port Authority should monitor congestion pricing effects as a **network redistribution problem**, not only as a facility-by-facility volume problem.

The **first** step should be to track traffic share by facility every month. Instead of only reviewing whether total traffic increased or decreased, Port Authority should monitor whether a facility is taking on a larger share of total system traffic. This is important because congestion may not disappear; it may shift from one facility to another.

Second, facilities gaining share in 2025 should be monitored for secondary congestion risk. George Washington Bridge Upper, Goethals Bridge, George Washington Bridge Lower, and Bayonne Bridge gained traffic share. These facilities should be reviewed for signs of increased queuing, slower movement, higher incident sensitivity, or rising violation pressure.

Third, facilities losing share should also be studied, not ignored. Lincoln Tunnel, Holland Tunnel, and Outerbridge Crossing showed traffic-share declines. The question for management should be whether those declines reflect successful congestion reduction, toll avoidance, route substitution, temporary partial-year effects, or other external conditions. This requires continued monitoring rather than a one-time comparison.

Fourth, Port Authority should create a before-and-after facility dashboard for 2025 and later years. The dashboard should include total volume, traffic share, change in share, violation count, violation rate, day-of-week distribution, and weather overlays. This would allow management to detect whether traffic redistribution is temporary or sustained.

Fifth, any pricing-related analysis should be reviewed at the system level. If pricing reduces traffic pressure at one location but increases pressure at another, the operational burden has not been removed; it has been redistributed. Port Authority should therefore evaluate whether policy changes reduce total congestion or simply transfer congestion to different crossings.

Project Question 5

Forecast the usage of facilities beyond 2025 to the maximum year possible

Business Purpose

Port Authority wanted to forecast future bridge and tunnel usage beyond 2025. This question is different from the earlier questions because it requires projecting future values based on historical patterns. Therefore, it was treated as a time-series forecasting problem.

Forecasting Approach

An initial simple forecasting approach using highly aggregated total traffic produced a flat forecast. This was not suitable because it lost important facility-level variation. A single system-wide forecast does not reflect how Port Authority operates, because each facility has different usage patterns, seasonal behavior, and structural demand.

The approach was therefore refined to **facility-level time-series forecasting**. Each facility was treated as a separate time series, and monthly aggregation was used. Monthly data was selected because it is detailed enough to preserve seasonality while being stable enough for forecasting.

The forecasting structure was:

Element	Forecasting Design
Time variable	Monthly date
Target variable	Total traffic
Series identifier	Facility
Forecast level	Facility-level
Reason for monthly grain	Preserves seasonality while reducing daily noise

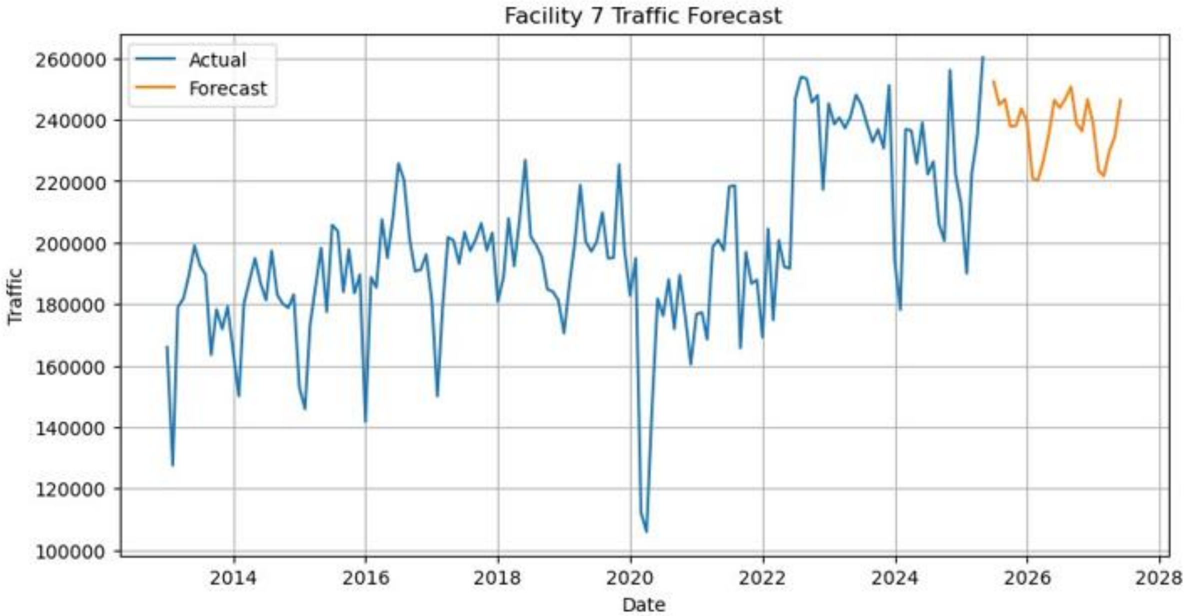
This approach is better aligned with Port Authority operations because decisions about staffing, facility monitoring, congestion, and infrastructure planning are made at the facility level, not only at the system-wide level.

Final Answer

The forecast was designed to estimate future usage separately for each facility rather than relying on one total system forecast. This is important because historical analysis showed that traffic demand is not evenly distributed. George Washington Bridge-related facilities, Lincoln Tunnel, Holland Tunnel, Goethals Bridge, Bayonne Bridge, and Outerbridge Crossing each show different volume levels, seasonal patterns, and operational behavior.

The final forecasting approach therefore supports facility-specific planning. It allows Port Authority to identify which facilities are expected to remain high-demand, which may face growth pressure, and which may require closer monitoring for future congestion risk.

Because the provided report text confirms the forecasting design but does not include the complete final forecast table in the visible uploaded section, the final Word report should include the actual forecast chart or forecast table from the notebook after this subsection. The explanation below is written so that the forecast visual can be inserted without creating unsupported numeric claims.



Facility-Level Monthly Forecast Beyond 2025

Forecast:

series_id	1	2	4	5 \
2025-06-30	129907.633630	158640.585449	37255.451041	159375.398826
2025-07-31	127159.608739	155975.731755	36172.626186	158304.097440
2025-08-31	130065.598744	159289.791819	34889.463766	158556.537620
2025-09-30	128074.958290	157021.291623	33587.308645	154984.798122
2025-10-31	128845.686414	159005.472049	33243.577174	152931.171009

series_id	6	7	8	9
2025-06-30	127212.700714	252364.425355	19718.181032	192585.782842
2025-07-31	126996.953964	244834.316278	17727.393179	193660.268700
2025-08-31	128868.202507	246637.061544	20096.056955	188084.764179
2025-09-30	124016.322592	237847.282768	19195.903545	182563.438274
2025-10-31	122800.363379	237948.419772	20125.553866	186402.323470

Recommendation for Port Authority

Port Authority should use the forecast as a **facility-level planning input**, not as a single system-wide traffic estimate.

First, the forecast should be reviewed separately for each facility. High-volume facilities such as George Washington Bridge Upper, George Washington Bridge Lower, and Lincoln

Tunnel should receive individual forecast monitoring because small percentage changes at those facilities can represent large absolute changes in vehicle volume.

Second, forecast outputs should be connected to operational planning. If a facility is projected to increase during specific months, Port Authority should review staffing, lane availability, incident-response readiness, and maintenance scheduling before those months occur.

Third, forecast results should be refreshed regularly. Traffic patterns have shown structural disruption around 2020 and redistribution in 2025, so a static forecast should not be treated as permanent. The forecast should be updated as new monthly data becomes available, especially after pricing changes, major policy changes, major weather events, construction changes, or regional travel disruptions.

Fourth, forecast monitoring should include forecast error tracking. After actual traffic data becomes available, Port Authority should compare actual usage against forecasted usage by facility. If forecast errors increase at a specific facility, that may indicate a new structural shift or an external factor not captured in the historical pattern.

Fifth, forecast results should be used with scenario planning. For example, Port Authority can maintain a baseline forecast, a high-demand scenario, and a disruption scenario. This would help management prepare for both normal growth and abnormal events.

Overall Model Development Conclusion

The model development phase moved the project from data preparation into business answers.

For Question 1, regression modeling showed that facility, day of week, and weather variables were the leading usable drivers of facility usage once direct components of total traffic were excluded. Facility was the strongest driver, confirming that Port Authority usage patterns are highly location-specific.

For Question 2, descriptive analysis showed that toll violations are large in scale, with more than **8 million violations**, and highly concentrated in George Washington Bridge Lower, George Washington Bridge Upper, and Lincoln Tunnel.

For Question 3, traffic was shown to follow clear yearly, monthly, weekly, and facility-specific patterns. Traffic peaks during summer months, remains strong on weekends, and is driven primarily by passenger vehicles.

For Question 4, the 2025 analysis showed that traffic share shifted across facilities. George Washington Bridge Upper, Goethals Bridge, George Washington Bridge Lower, and Bayonne Bridge gained share, while Lincoln Tunnel, Holland Tunnel, and Outerbridge Crossing declined. The classification model confirmed that high-traffic conditions are strongly facility-driven.

For Question 5, the forecasting approach was refined from a system-wide forecast to facility-level monthly forecasting because facility-level forecasting better reflects Port Authority's operational reality.

The overall conclusion is that Port Authority traffic is structured, facility-specific, seasonal, and behaviorally sensitive. The strongest management value comes from monitoring traffic at the facility level, tracking violations by high-risk locations and seasons, preparing for summer and weekend demand, reviewing 2025 redistribution effects, and refreshing forecasts regularly as new data becomes available.

Project Question 6:

Develop the model using Python code with Jupyter Notebook and Anaconda

Final Answer

The modeling and forecasting work was implemented using Python in Jupyter Notebook within the Anaconda environment. H2O AutoML was used for regression and classification modeling, while the forecasting phase was structured using time-series data preparation and facility-level monthly aggregation. The work was organized through a notebook workflow that followed the project phases: data audit, exploratory analysis, data cleaning and integration, sampling, model development, and forecasting.

For management reporting, the code itself is not included in the final report because the report is intended to explain the decisions, findings, and recommendations. However, the notebook files provide the technical implementation and reproducible workflow behind the results.

Python Model Implementation Appendix

After completing the AutoML-based model development phase, the selected modeling logic was also implemented using standard Python libraries. This appendix was created to demonstrate that the modeling approach is reproducible, interpretable, and not dependent only on AutoML outputs.AutoML was useful because it tested multiple candidate models and helped identify strong-performing approaches objectively. However, for a project intended to support real business decision-making, it is important to show that the models can also be understood and rebuilt using traditional Python-based methods. This makes the project more transparent and gives Port Authority confidence that the modeling process is not a black box.

The Python implementation focused on three modeling tasks:

Model Type	Business Purpose
Regression Model	Predict total traffic volume
Classification Model	Identify high-traffic or congestion-prone conditions
Time-Series Model	Forecast future traffic demand

Each model was developed for a different business purpose. The regression model answered how traffic volume can be estimated from external and operational factors. The classification model helped identify whether a given facility-date is likely to represent a high-traffic condition. The time-series model supported future planning by forecasting traffic demand beyond the observed period.

Regression Model: Predicting Traffic Volume

The first Python-based model was a regression model designed to predict total traffic volume. This model was appropriate because the target variable, **TOTAL traffic**, is a continuous numeric value.

The purpose of this model was not only to predict traffic, but also to confirm which factors help explain traffic volume after removing variables that would cause data leakage. Variables such as cash payments, E-ZPass payments, violations, autos, trucks, and buses were excluded because they directly make up or strongly overlap with total traffic. Including those variables would have allowed the model to reconstruct total traffic mechanically instead of learning meaningful explanatory patterns.

The model used facility, day of week, weather variables, passenger volume, and bus volume as predictors. This allowed the model to focus on structural and external drivers rather than direct components of the target variable.

A Gradient Boosting model was used because this type of model can capture non-linear relationships between predictors and traffic volume. This is important because traffic behavior is rarely linear. Facility type, day-of-week behavior, weather, and passenger movement may interact in different ways across time and location.

The regression model produced the following results:

Metric	Result
RMSE	4,623.70
R^2 Score	0.9386

The R^2 value of approximately **0.94** indicates that the model explains about **94% of the variation in traffic volume**. This is a strong result for real-world transportation data, especially because direct traffic components were excluded.

The RMSE of approximately **4,624 vehicles** means that, on average, the model's prediction error is within a few thousand vehicles. Given that facility-level daily traffic can vary widely across facilities and dates, this level of error is reasonable for planning and pattern-identification purposes.

Interpretation

The regression model confirms that traffic volume can be predicted effectively using facility, day-of-week, weather, and transit-related variables. The model's strong performance supports the earlier finding that traffic is structured and explainable rather than random.

The actual-versus-predicted chart showed that predicted traffic values closely followed actual traffic values. Most points aligned near the diagonal reference line, which means the model captured the overall traffic pattern well.

The residual analysis showed that prediction errors were centered around zero, with no strong systematic pattern. This suggests that the model was not consistently overpredicting or underpredicting traffic. A slight increase in error spread at higher traffic levels was observed, which is expected in transportation data because peak traffic days are naturally more volatile.

Business Meaning

For Port Authority, this model shows that traffic volume can be estimated using a combination of facility, time, weather, and supporting demand indicators. This can support planning decisions such as expected facility demand, peak-period preparation, and future model-based monitoring.

The model should not be used as the only basis for operational decisions, but it provides a strong analytical foundation for understanding which conditions are associated with higher or lower traffic.

Classification Model: Identifying High-Traffic Conditions

The second Python-based model was a classification model designed to identify whether a given facility-date represents a high-traffic condition.

This model was different from the regression model. Instead of predicting the exact number of vehicles, it classified traffic into two categories:

Class	**Meaning**
Normal Traffic	Traffic below the high-traffic threshold
High Traffic	Traffic at or above the high-traffic threshold

High traffic was defined using the **75th percentile of total traffic**. This means the model focused on the top quarter of traffic conditions rather than using an arbitrary cutoff. This is a defensible approach because it defines congestion-prone conditions based on the actual distribution of the data.

A Random Forest classifier was used because it handles non-linear relationships and interactions well. This is useful for transportation analysis because high-traffic conditions may depend on facility, day of week, weather, and other variables at the same time.

The classification model produced the following results:

Metric	Result
Accuracy	91.94%

The classification report showed strong performance for both classes. The model performed especially well in identifying normal traffic conditions, with precision and recall around **0.95**. It also performed well in identifying high-traffic conditions, with precision and recall around **0.84**.

Class	Precision	Recall	F1-Score
Normal Traffic	0.95	0.95	0.95
High Traffic	0.84	0.84	0.84
Overall Accuracy			0.92

Interpretation

The classification model successfully identified high-traffic conditions with strong accuracy. The confusion matrix showed that most normal and high-traffic cases were correctly classified, and misclassification was relatively limited.

The feature importance analysis showed that facility-related variables, especially George Washington Bridge-related facilities, were among the strongest predictors of high traffic. Weather variables such as temperature also contributed, but to a smaller degree.

This confirms one of the strongest findings of the project: congestion and high-traffic conditions are primarily facility-driven. External conditions matter, but the facility itself is the dominant factor.

Business Meaning

For Port Authority, the classification model can be used as an early-warning framework for high-traffic conditions. Instead of only reacting after congestion occurs, management can monitor facility, day-of-week, weather, and passenger/bus indicators to identify dates that are more likely to experience high demand.

This type of model can support:

- Peak-period planning
- Facility-level monitoring

- Congestion readiness
- Staffing decisions
- Enforcement scheduling
- Dashboard-based alerts

The model is not a replacement for real-time operations, but it provides a decision-support layer that helps identify when and where high traffic is more likely.

Time-Series Model: Forecasting Future Traffic Demand

The third Python-based model was a time-series forecasting model. This model was developed to estimate future traffic demand beyond the observed period.

Traffic data is time-dependent, meaning current and future traffic levels are influenced by historical patterns, seasonality, and recurring demand cycles. For this reason, a time-series model was appropriate.

A SARIMA model was used because it can capture both trend and seasonality. This is important because earlier analysis showed that Port Authority traffic follows seasonal patterns, with higher traffic during warmer months and lower traffic during colder months.

Facility 7, which corresponds to **George Washington Bridge Upper**, was selected for the Python forecasting implementation because it is one of the highest-traffic facilities in the dataset. High-volume facilities are operationally important because even small changes in demand can represent large absolute changes in vehicle volume.

The model forecasted traffic for the next 12 months. The forecasted values remained within a stable range, indicating that future traffic demand for the selected facility was not expected to show extreme growth or extreme decline within the forecast window.

Interpretation

The forecast showed that traffic demand is expected to remain broadly stable while continuing to follow seasonal movement. This means future traffic is likely to fluctuate by month, but the model did not indicate an extreme upward or downward shift.

This supports the broader conclusion that Port Authority traffic is structured and seasonal. Future demand planning should therefore continue to account for monthly patterns, facility-level differences, and recurring high-demand periods.

Business Meaning

For Port Authority, forecasting is useful because it supports proactive planning. If a facility is expected to experience higher traffic during certain future months, operational teams can prepare in advance through staffing, maintenance scheduling, lane planning, incident-response readiness, and enforcement coordination.

The forecast should be updated regularly as new traffic data becomes available. This is especially important because earlier analysis showed structural changes around 2020 and traffic-share shifts in 2025. A forecast should not be treated as permanent; it should be refreshed as new conditions emerge.

Why the Python Appendix Matters

This appendix strengthens the overall project because it shows that the modeling process is reproducible outside of AutoML. AutoML helped identify strong model candidates, but Python implementation confirmed that the modeling logic can be rebuilt, evaluated, visualized, and interpreted using standard methods.

The Python models also provide different types of decision support:

Model	What It Helps Port Authority Understand
Regression	How traffic volume changes based on facility, time, weather, and demand indicators
Classification	Which conditions are associated with high traffic or congestion-prone periods
Time Series	How future facility traffic may evolve over time

Together, these models support a more complete view of Port Authority traffic behavior. They show not only what happened historically, but also how traffic can be predicted, classified, and forecasted for planning purposes.

Power BI Dashboard Development Plan

After completing the analysis and modeling phases, the next step is to build a Power BI dashboard. The dashboard is required as part of the final project deliverables and will serve as the interactive management view of the project.

The report explains the logic, methods, findings, and recommendations in detail. The Power BI dashboard will allow Port Authority stakeholders to explore those findings visually and interactively.

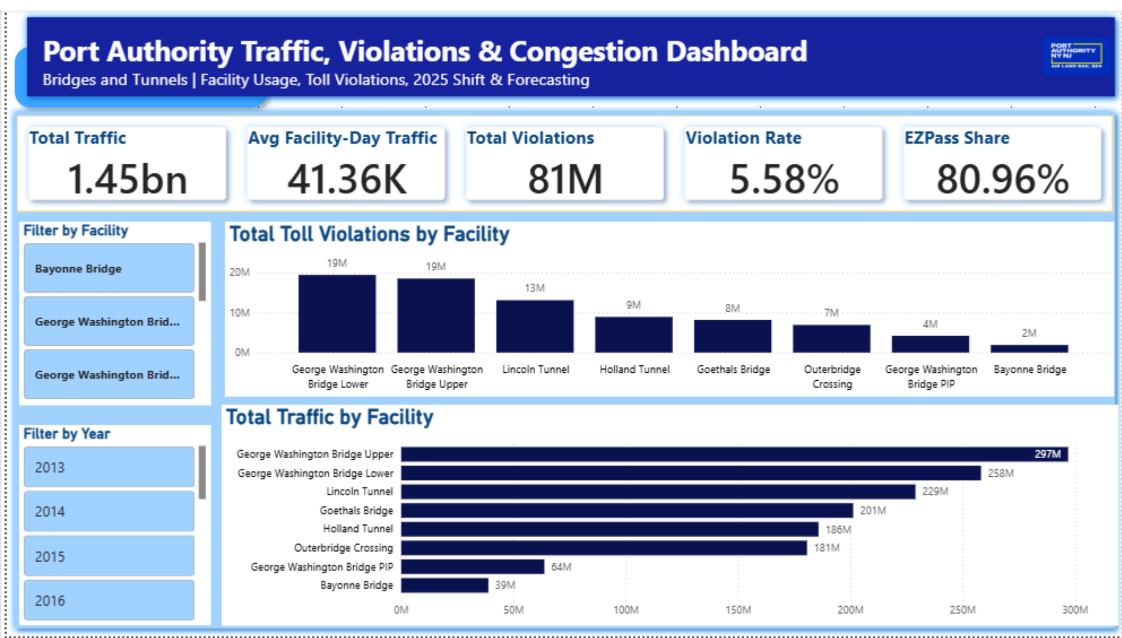
Why a Power BI Dashboard Is Needed

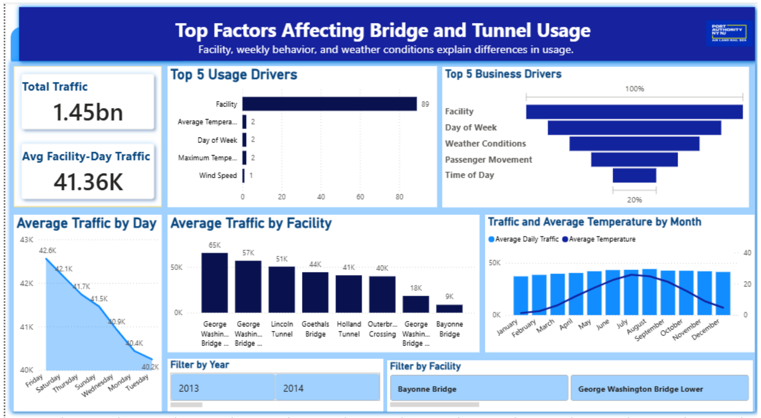
The analysis contains multiple layers: facility usage, toll violations, busiest periods, seasonal patterns, 2025 traffic shifts, and forecasted demand. A written report is useful for explanation, but a dashboard is better for monitoring and decision-making.

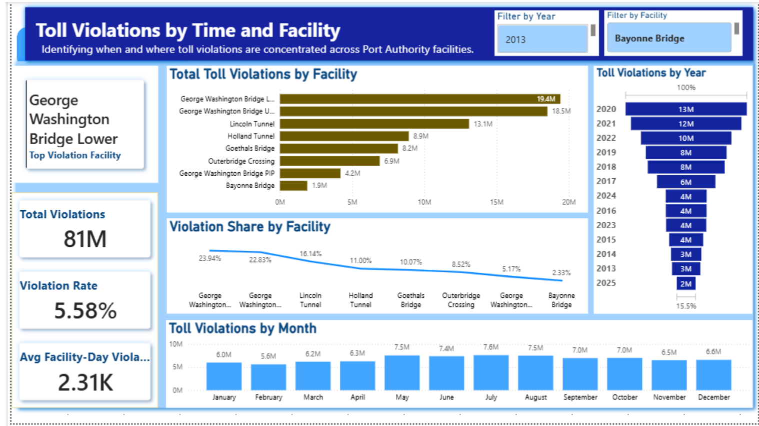
The dashboard will allow users to filter by facility, time period, year, month, and day of week. This is important because Port Authority decisions are facility-specific and time-sensitive. A static chart in the report can show one view, but Power BI allows users to explore multiple views without rebuilding the analysis.

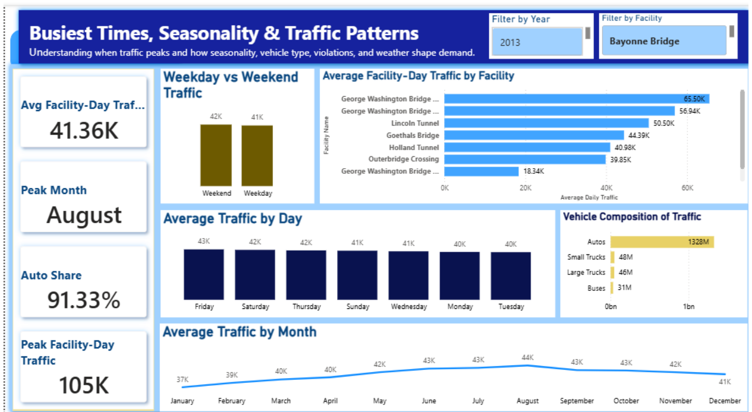
The dashboard will also make the final results easier to communicate. Port Authority stakeholders can quickly see where traffic is highest, where violations are concentrated, how demand changes over time, and which facilities require closer attention.

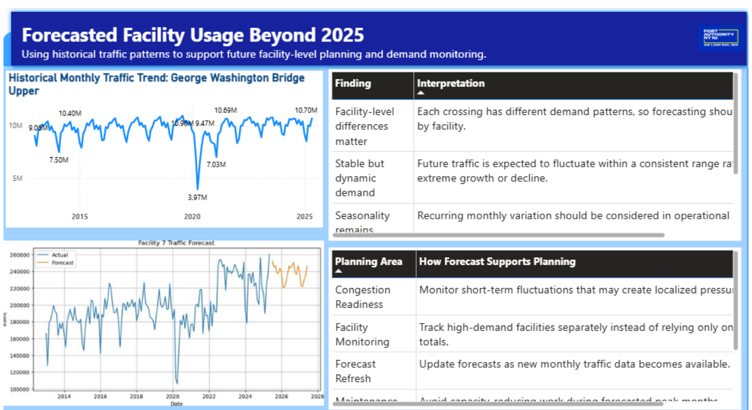
Power BI Dashboard Preview

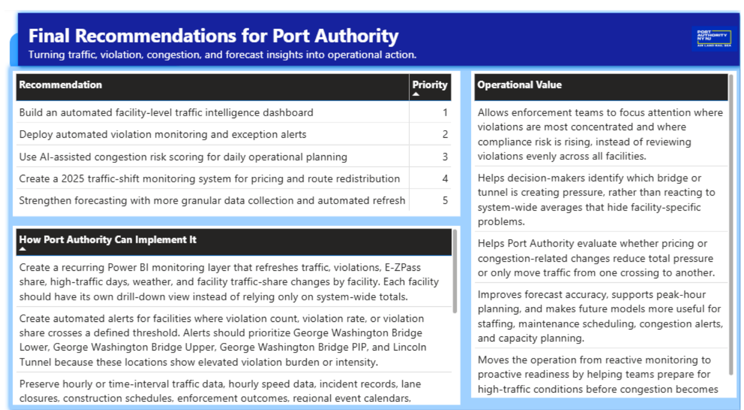












Click to [Review the interactive Power BI dashboard here]

Final Project Conclusion

This project analyzed **Port Authority bridge and tunnel traffic** to understand **usage drivers, toll violations, busiest travel periods, congestion-related traffic shifts, and future facility demand**.

The project began by auditing and preparing the raw datasets. The data was then cleaned, aligned to a **Facility × Date grain**, integrated with weather and transit-related variables, and sampled using a **stratified approach** that preserved facility and monthly time coverage. This created a reliable modeling dataset for answering the project questions.

The analysis showed that **Port Authority traffic is structured, not random**. **Facility location is the strongest driver of traffic usage**. Day of week and weather conditions also influence traffic, but **facility-level differences dominate the system**. This means Port Authority should monitor demand at the **facility level** rather than relying only on system-wide averages.

Toll violations were found to be large in scale and highly concentrated. More than **8 million violations** were identified, with **George Washington Bridge Lower and George Washington Bridge Upper together accounting for nearly half of total violations**. This suggests that violation management should be targeted by **facility and season** rather than treated as a uniform system-wide issue.

Traffic patterns also showed **clear seasonality**. Demand rises through spring, peaks during the **summer months**, and remains strong on **weekends**. **Passenger vehicles are the main contributor to traffic variation**, while weather plays a supporting role by reinforcing seasonal patterns and affecting travel during adverse conditions.

The **2025 analysis showed that traffic share shifted across facilities**. Some facilities gained traffic share while others declined, indicating that congestion or pricing-related conditions may **redistribute traffic rather than eliminate pressure entirely**. This finding is important because Port Authority should monitor congestion as a **network-wide issue**. A reduction at one facility may create increased pressure at another.

The forecasting phase showed that **future planning should be conducted at the facility level**. A system-wide forecast is not enough because each facility has different usage patterns and operational importance. **Facility-level forecasting** can help Port Authority prepare for future demand, maintenance windows, staffing needs, and congestion risk.

The Python appendix confirmed that the modeling approach is **reproducible and interpretable**. Regression, classification, and time-series models were implemented

using standard Python methods, supporting the AutoML findings and showing that the project's conclusions are **not dependent on black-box tools alone**.
Overall, this project provides Port Authority with a **decision-support framework** for monitoring facility usage, managing toll violations, preparing for seasonal and weekend traffic, evaluating 2025 traffic redistribution, and planning future demand. The final Power BI dashboard will translate these findings into an **interactive tool** that allows stakeholders to explore the results by facility, time period, and project question.

Appendix: Supporting Project Files

The supporting notebooks, PDF exports, final dataset, and dashboard files are available through the GitHub repository linked below. These files document the complete analytical workflow used in this project, including data audit, exploratory analysis, cleaning and integration, sampling, model development, Python implementation, and dashboard preparation.

File	**Description**	**Link**
01 Data Audit	Initial data structure review and quality checks	Open
02 Descriptive Analysis and EDA	Exploratory analysis of traffic, violations, facilities, and time patterns	Open
03 Data Cleaning and Integration	Cleaning, grain alignment, and final master dataset creation	Open
04 Sampling	Representative sample creation for model development	Open
05 Model Development	H2O AutoML, project-question analysis, and forecasting work	Open

06 Python Model Implementation	Reproducible Python implementation of selected models	Open
Power BI Dashboard	Interactive dashboard for final project findings	Open
Final Dataset	Cleaned integrated dataset used for reporting and dashboarding	Open